ZS-A27

The Z-Spin Vacuum-Energy Scale: A Dimensional No-Go for the Cosmological Constant, the Four-Root Map, and the Closure Frontier

Kenny Kang · Z-Spin Collaboration · June 2026 · Theme: Astrophysics · **ZS-A27 v2.3**
Diagnostic checks (v2.3): channel-(ii) transmutation computation (§7.1: blind $\rho\Lambda^{1/4} = 2.48\times10^{-5}$ eV; the 112/120-orders figure is a post-unsealing comparison, not a blind output); the odd determinant–torsion identification is HYPOTHESIS (§8.1, with an even-dimension vanishing concern); the Selective Sequestering Compatibility Theorem is registered as the top OPEN problem (§8.2); multiplicity-2 audit (§5.3) + corrective / mass-gap / integration suites | Zero fitted parameters (candidate-action coefficients OPEN) | (A = 35/437, Q = 11, dim Z = 2) LOCKED

Status of the central claim. The absolute vacuum-energy scale B3 is **OPEN** — not closed, and not provably impossible. This is the **public release (v2.3)** of the internal v1.0–v1.10 line: the older half-line/Weyl-clock analysis is a limited Route-A diagnostic and the calculations it leaned on are retracted (Appendix B), as is the v1.7 dark-energy lever (Appendix A.5). The forward program is four roots — a product-trace bridge (DERIVED-CONDITIONAL), even-sector sequestering of the bare offset (OPEN), an odd four-form residual (HYPOTHESIS-strong, with its multiplicity-2 step reframed as a rank test in §5.3), and a blind scale gate (OPEN) — sitting on top of the §7 scale-generation no-go (DERIVED, restricted): A, Q and the dimensionless invariants alone cannot fix the scale, though the one locked dimensionful object M̄_P together with an action-fixed mechanism can. R0 now carries an external Kaloper–Padilla sequestering lift (§4).

§0. Abstract

The cosmological-constant value problem appears in Z-Spin Cosmology as the **absolute vacuum-energy scale B3**: the additive offset of the vacuum energy, which the corpus's relative spectral and modular tools (ZS-A26) cannot see because it is a central shift. We give the corrected map for B3. **First**, we state precisely the scopes in which closing B3 is a **CONDITIONAL NO-GO** — within the locked local master action, within relative-only tools, for the naive half-line canonical realization, and for a constant Ohmic freeze — and we are explicit that an *absolute* no-go is *not* established (the v1.3 "certified-unclosable" claim was withdrawn). **Second**, we replace the retracted register→metric no-go (C1) with the almost-commutative **product-trace bridge C1′** (Theorem 1): an imported spacetime spectral triple tensored with the finite Z-register carries a factorized trace that couples the register to the gravitational local trace without lowering the metric dimension; the algebraic half is already proved in the corpus (ZS-F23.4). This is metric *coupling*, not emergence, and it removes the need to input the global four-volume. **Third**, we replace the retracted Schwinger–Keldysh freeze (R3) with the **four-form residual programme M1** (Theorem 2): a Z_2-odd four-form linearly coupled to a dimension-4 electroweak operator yields a positive, exactly constant residual $\rho_\Lambda = \chi_Z(O_{EW}^{(-)})^2/(2\bar{M}_P^4)$ with no friction or freeze time — conditional on four results the corpus does not yet have. **Fourth**, since M1's residual sits on top of the bare offset c_0 ($\rho_{grav} = c_0 + \rho_{M1}$), the map needs a fourth root **R0** — even-sector sequestering

that removes every constant vacuum shift (the old cosmological-constant problem) — so the corrected structure is the four roots **{C1′, R0, M1, S1}**, of which only C1′ is closed (conditionally). **Fifth and centrally**, we prove a restricted no-go (§7, **A–Q-Only Scale-Generation No-Go**, DERIVED): the dimensionless data — A, Q, the relaxation rates, the embedding weights — cannot, by Buckingham-π, fix the dimensionful ρ_Λ on their own; closing B3 needs at least one dimensionful normalization (the one locked dimensionful object is $\bar{M}_P$) or a dynamical scale-generating mechanism. *We explicitly weaken the v2.0 over-statement*: the full corpus is not "purely dimensionless" (it carries $\bar{M}_P$, v, Λ_{QCD}), and the absolute scale is a hierarchy/selector problem, not a dimensional one. The currently-known routes are a **non-exhaustive taxonomy** (HYPOTHESIS), not an exhaustive trichotomy: (i) an exponent *fit to the target* is numerology (barred), but an exponent *fixed by the action/topology* (dimensional transmutation $\Lambda_- = \bar{M}_P \exp[-8\pi^2/b_0 g_-^2]$) is *legitimate* nonperturbative physics; (ii) a new confining sector; (iii) the observed Hubble scale (dynamical DE) — with full action extensions, the SM running sector, gravity anomalies, topological quantization, and nonperturbative saddles all left open. The corpus supplies the dimensionless coefficient and the M1 coupling structure; it does not, from A and Q alone, supply the scale. **We then compute that one no-observation route (channel ii, §7.1).** Riding the corpus's DERIVED Higgs-VEV transmutation blind (no v_{now}, no epoch, no observed ρ_Λ) yields a small blind benchmark $\rho_\Lambda^{1/4} = 2.48\times10^{-5}$ eV that lands *within ~8 of the 120 orders* the additive cutoff^4 estimate misses — a **post-unsealing comparison**, not a validation of the M1 mechanism — but **90× short** in scale, the closing ×9.5 being the documented S1 trap. The odd spectral invariant that would force the residual exponent is **one necessary-not-sufficient S1 subgate** (its identification with a Z_2-odd Ray–Singer analytic torsion is a HYPOTHESIS facing a possible even-dimension vanishing, §8.1); the **top open problem** is instead the Selective Sequestering Compatibility Theorem binding R0 and M1 in one action (§8.2). A methodological note records that this channel-(ii) analysis originated in AI-assisted deep-search, so its internal consistency is not independent validation — external review (NCG specialist) and data (DESI, channel iii) are what would settle it. **Sixth**, we correct two over-reaches: the ZS-Q7 "gap" is an exact *dimensionless Liouvillian relaxation* gap, not a physical mass gap (so "decoupling answered" and "TeV/MeV scale" claims are withdrawn); and the v1.7 dark-energy lever is retracted — the ghost scale $\Lambda_Z \approx 25.6$ MeV was back-solved from observation, the two scalings $\rho \propto \Lambda^3 H$ and $\rho \propto \Lambda^8/\bar{M}_P^4$ cannot be mixed, and the ghost $w_0 = -0.763$ is a toy discriminator, not a confirmed prediction. We conclude that **B3 remains OPEN** — the A–Q-only inputs *underdetermine* it (§7), and the known routes to closure are classified but non-exhaustive; both a CLOSED declaration and an absolute no-go are explicitly declined. The constructive programme ($H_{J=-}^{0,4} \to G_- \to \Lambda_- \to \chi_- \to \rho_\Lambda^{pred}$, under a blind protocol) is deferred to ZS-A28. The checks are reproduced by *verify_v23.py*, *verify_v18.py* and the earlier suites; they are consistency and diagnostic checks, not physical proofs. No fitted parameter is introduced; 3πv, 9v, and the back-solved TeV/MeV targets are REJECTED-COINCIDENCE.

Condition (ii), sharpened (§5.3). We sharpen condition (ii). The J_Z-odd dimension-4 cohomology realized in ZS-M27 is *two*-dimensional (the two ramified-prime characters χ_{-3}, χ_{-11}), so M1's uniqueness is a genuine multiplicity-2; the determinant-line reduction is **parity-obstructed** (the top exterior power is J_Z-even, while M1 needs J_Z-odd), whereas the corpus-admissible cross-coupled quadratic invariant $\rho_\Lambda = \Omega_-^T K_- \Omega_-$ removes the need for any generator selection and converts condition (ii) into a finite **rank test** on the ZS-M33 Z-mediated kernel — a reframing, not a closure.

Consolidated status of the four roots (v2.3). C1′, R0, M1, and S1 sit above the restricted A–Q-only no-go (§7); B3 overall remains OPEN.

Root / result	Statement	Status
C1′ — product-trace bridge (§3)	Almost-commutative spectral triple couples the Z-register to the gravitational local trace; algebraic half PROVEN (ZS-F23.4). Metric coupling, not emergence.	**DERIVED-CONDITIONAL**
R0 — even sequestering (§4)	Removes every constant vacuum shift (c_0); Kaloper–Padilla sequestering supplies an external four-form / 4-volume realization with no new local DOF.	**OPEN** (DERIVED-COND. lift via KP)
M1 — odd four-form residual (§5)	Z_2-odd four-form → positive, exactly constant ρ_Λ; four conditions (§5.1).	**HYPOTHESIS-strong**
M1 condition (ii) — §5.3	Determinant-line route NON-CLAIM (parity); cross-coupled quadratic invariant viable; B3-2 recast as a rank test on the ZS-M33 kernel; χ_{-3}/χ_{-11} legs DERIVED (M30) / PROVEN (M22).	**REFRAMED**; rank OPEN
S1 — blind absolute scale (§6)	KV-vs-Zhitnitsky scaling fork; 9v, 3πv, and back-solved TeV/MeV targets REJECTED-COINCIDENCE.	**OPEN**
A–Q-only scale no-go (§7)	A, Q + dimensionless invariants alone cannot fix the dimensionful ρ_Λ (Buckingham-π); known routes are a non-exhaustive taxonomy.	**DERIVED** (restricted)
Channel (ii), §7.1 — transmutation frame	Blind transmutation → 2.48×10^{-5} eV (the 112/120 is a post-unsealing comparison, not a blind output); 90× short; the odd invariant is one necessary-not-sufficient S1 subgate, its Ray–Singer identification a HYPOTHESIS (even-dimension vanishing concern).	**FRAME HYP.**; closure OPEN
Selective Sequestering Compatibility, §8.2	R0+M1 in one action: $\partial\Delta V/\partial G = 0$ while $\partial\rho_{M1}/\partial G \neq 0$. The top open problem — without it a correct odd invariant still does not close B3.	**OPEN** (top priority)
B3 — absolute vacuum scale	The cosmological-constant value; underdetermined by A–Q-only inputs; both a CLOSED declaration and an absolute no-go declined.	**OPEN**

Epistemic Status Legend

STATUS	DEFINITION
PROVEN	Established by rigorous internal proof or a cited external theorem.
DERIVED	Follows from PROVEN results by explicit derivation.
DERIVED-CONDITIONAL	Follows rigorously once a stated import or assumption is granted.
HYPOTHESIS-strong	Multiple independent structural anchors; explicit promotion path documented.
CONDITIONAL NO-GO	A proven impossibility under stated assumptions; explicitly not absolute.
OPEN	Recognized gap, honestly registered for future work.
RELATIVE-BLIND	A correct statement about a relative quantity, insensitive to the offset c_0 (ZS-A26).
REJECTED-COINCIDENCE	A numerical near-match found after seeing the target; barred by anti-numerology.
RETRACTED	A claim withdrawn after audit; statement and reason kept on the record (Appendix B).
LOCKED	Immutable upstream constant; no downstream paper may modify.

§1. Introduction: the absolute vacuum scale as a central shift

Z-Spin Cosmology fixes essentially every dimensionless ratio from the geometric impedance **A = 35/437** = (5/19)(7/23) and the eleven-slot register **Q = 11** = (dim Z, dim X, dim Y) = (2, 3, 6). The one quantity that resists this is the *absolute* scale of the vacuum energy — the cosmological-constant value, here called **B3**. ZS-A26 isolated the reason: the corpus's working tools are *relative* — determinant ratios, normalized traces, modular ratios, transition probabilities, action differences — and an additive offset c_0 of the vacuum energy is a *central shift* to which every such quantity is blind (RELATIVE-BLIND). Fixing B3 therefore requires an object that is sensitive to the offset itself.

ZS-A27 has pursued this across several versions. The earliest (v1.0–v1.1) framed it as a half-line conjugate eigenvalue problem and a finite Weyl-clock; v1.2–v1.3 attached three concrete calculations and, in v1.3, over-reached to a "certified-unclosable" verdict. The v1.4 audit retracted that verdict and the three calculations behind it. **This is the public release (v2.3) of the corrected map**. The retracted half-line/Weyl-clock material and the withdrawn calculations are summarized in Appendix B (Correction History) and are not part of the current argument. The main body below is the forward program only.

§2. The conditional-no-go boundary, stated precisely

It is important to say exactly where B3 *is* obstructed and where it is not. In each of the following scopes, closing B3 is a **CONDITIONAL NO-GO** — a proven impossibility under a stated restriction, not an absolute one.

Scope	Statement	Status
A. Locked local master action	With no global constraint, sequestering four-form, odd memory flux, or offset-sensitive boundary state, the additive offset remains an independent coupling.	**CONDITIONAL NO-GO**
B. Relative-only tools	Determinant / trace / modular ratios, transition probabilities, action differences, Weyl-clock branch spacing cannot see a central shift.	**CONDITIONAL NO-GO**
C. Route-A half-line realization	The specific realization $L^2(\mathbb{R}_+)$, $\Lambda = -i\partial_T$ fails by deficiency index (1,0); not a no-go for all unimodular quantizations.	**NO-GO** (this representation)
D. Constant Ohmic freeze	$\gamma(t) = \gamma_0 > 0$ gives $\int\gamma\, dt = \infty$ and $q \to 0$; a constant friction makes no permanent residual.	**NO-GO** (this mechanism)

What is NOT established. No proof exists that *every* diffeomorphism-invariant, gauge-invariant, radiatively stable, zero-free-parameter Z-Spin extension fails to produce a small positive residual. On the contrary, this paper exhibits an explicit candidate (M1). Declaring an absolute no-go now would repeat exactly the v1.3 over-reach that v1.4 corrected. **B3 is OPEN**; the scopes above bound it, they do not close it. The path that would close it as an absolute no-go is stated in §8.

§2.1 Why the offset is a central shift: the emergent-gravity certification

Scope B — that the corpus tools are blind to a central shift — has an external anchor worth restating (it was developed in the v1.4 audit and is restored here). In any theory where the Einstein equation arises *thermodynamically* rather than from a fundamental metric action — Jacobson's derivation from the Clausius relation on local horizons, and the entanglement-equilibrium refinement [11] — the cosmological constant enters as an **integration constant**, not as a Lagrangian coupling. The field equations fix the *derivative* structure of gravity; the additive vacuum term is unconstrained by them. This is the same conclusion A26 reached internally (the corpus tools are relative), now cross-validated from outside: B3 is genuinely a central-shift problem, which is *why* no determinant, trace, or modular ratio in the corpus can fix it. **Status: DERIVED** (external), supporting A26.

The connection to the observational lever. An integration constant can be realized two ways, and that is exactly the M1 fork: a fixed constant (the KV branch, $w = -1$) or a quantity slaved to the infrared state of the universe (the Zhitnitsky branch, w tracking H). The emergent-gravity picture says gravity does not *care* which — both are admissible — so the choice is handed to observation, $w(z)$ (§6.4). The certification and the dark-energy lever are two ends of one statement: the vacuum scale is a free integration constant whose realization is measured, not derived from the relative corpus alone.

§3. Root C1′: the almost-commutative product-trace bridge

The retracted C1 argued that a finite register cannot carry a gravitational four-volume trace because its spectral (metric) dimension is $p = 0$ while a metric has $p = 3$. That only re-proves ZS-A17 — a finite register *alone* has no Weyl growth — and it conflated the spatial $p = 3$ with the spacetime $p = 4$. It does not show that an *imported* metric and the finite register admit no trace bridge. The correct construction is the standard almost-commutative product of noncommutative geometry [1], the one that geometrizes the Standard Model [2].

Take the imported spacetime spectral triple $\mathbb{Z}_X = (C^\infty(M_{BCC}), L^2(M, S), D_X)$ and the finite register triple $\mathbb{Z}_Z = (\mathfrak{H}_Z, \mathbb{C}^{11}, D_Z)$ with $\mathfrak{H}_Z = M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ (ZS-Q11, PROVEN). The product $\mathbb{Z}_{phys} = \mathbb{Z}_X \otimes \mathbb{Z}_Z$ has $D_{phys}^2 = D_X^2 \otimes I + I \otimes D_Z^2$, so the heat trace factorizes exactly and the local trace splits:

$$\mathrm{Tr}\, e^{-tD\_phys^2} = (\mathrm{Tr}\, e^{-tD\_X^2})(\mathrm{Tr}\, e^{-tD\_Z^2}), \quad \tau_{phys}(f \otimes B) = \int_M d^4x \sqrt{-g} \cdot f(x) \cdot (1/Q)\, \mathrm{Tr}_{\mathbb{C}^{11}} B.$$

The finite factor $\mathrm{Tr}\, e^{-tD\_Z^2} \to Q$ as $t \to 0$ *multiplies* the continuous heat-trace coefficient; it does not lower the metric dimension ($p_{total} = p_X$, verified). The algebraic half of the bridge is already a corpus theorem: ZS-F23.4 (DERIVED) is a trace-preserving central embedding of $M_3 \oplus \mathbb{C} \oplus M_5$ into the hyperfinite II_1 factor, sending the sector projections to weights $(3, 2, 6)/11$.

Theorem 1 (Product-Trace Bridge, C1′). Given an imported spacetime spectral triple, the almost-commutative product with the finite Z-register carries a factorized trace $\tau_{phys} = \int\sqrt{-g}\,(\cdot) \otimes (1/Q)\mathrm{Tr}_Z$ coupling the register to the gravitational local trace without lowering the metric dimension; the register half is the ZS-F23.4 embedding. **Status: DERIVED-CONDITIONAL** on the metric import.

Consequence. C1′ is a metric *coupling* theorem, not metric *emergence* (which ZS-A17 forbids). Because the trace is local — weighted by $\sqrt{-g}$ pointwise — a *local* vacuum-energy density does not require the global four-volume N_4 as input. This corrects the v1.2 claim that importing the metric leaves the whole problem untouched: it leaves the *global* count untouched, but Route B's target is the local density.

§4. Root R0: even sequestering, and why M1 alone is not enough

A correction prompted by external review: the corrected map needs a **fourth root**. Even if M1 (§5) generates a small positive residual, that residual sits *on top of* the bare vacuum offset c_0 unless an independent mechanism removes c_0. Schematically, the gravitating density is

$\rho_{grav} = c_0 + \rho_{M1}$ — and with c_0 a free Planck-scale offset, B3 is untouched no matter how small ρ_{M1} is.

So the absolute scale has **four** roots, not three: **C1′** (the metric–register product trace, §3), **R0** (even-sector sequestering that removes every constant vacuum shift), **M1** (the odd topological residual, §5), and **S1** (the dimensionful scale and matching coefficient, §6). R0 is the *old* cosmological-constant problem — radiative stability of the offset — and it is logically prior to M1: without R0 the residual is meaningless.

What R0 requires. In a sequestering action $S_{seq}^{(+)}$ built from Z_2-even four-forms, the physical Einstein tensor must be independent of every constant vacuum shift,

$$\partial G_{\mu\nu}^{phys}/\partial(\Delta V_{bare}) = \partial G_{\mu\nu}^{phys}/\partial(\Delta V_{EW}) = \partial G_{\mu\nu}^{phys}/\partial(\Delta V_{QCD}) = \partial G_{\mu\nu}^{phys}/\partial(\Delta V_{grav}) = 0,$$

covering matter one-loop, electroweak and QCD condensate shifts, and graviton-loop counterterms. External local-sequestering and its graviton-loop extension (Kaloper–Padilla [6]) show this can be arranged, but only within a chosen action, and standard sequestering *fixes no residual value* — it removes the offset and leaves the small piece to the odd sector. **Status: OPEN.** Verified as a structural point by *verify_v17.py* (R0): with c_0 present, ρ_{grav} is dominated by the offset, so M1 cannot close B3 on its own.

Immediate-rejection condition: if a single constant $\Delta V\, g_{\mu\nu}$ survives, B3 does not close.

A sequestering lift (external review, Step-2.5). Kaloper–Padilla vacuum-energy sequestering [6] is exactly a four-form / 4-volume mechanism that removes c_0 with no new local propagating degree of freedom: it makes the matter-sector dimensional parameters functionals of the 4-volume $\int d^4x\sqrt{g}$ and fixes the bare term by the global constraint $\lambda = \frac{1}{4}\langle T^{\mu}{}_{\mu}\rangle$, cancelling all matter loops and phase-transition shifts and (with the graviton-loop extension [6]) virtual-graviton contributions, evading Weinberg's no-go by acting only in the global / infinite-wavelength limit. The corpus already carries the ingredients to host it: the 4-volume clock T_4 / ν_{now} (ZS-A24, ZS-A26), the Z_2-even four-form programme of this section, and $\rho_Z = 0$ (ZS-F9, PROVEN: the Z-sector does not gravitate its own vacuum energy) — the Z-Spin form of Volovik's "equilibrium vacuum does not gravitate." Identifying T_4 with the sequestering 4-volume reads A26's relational law $\rho_\Lambda/\bar{M}^4 \propto 1/\sqrt{N_4}$ as the historic-average constraint, and $\rho_Z = 0$ as the native anchor for c_0 removal.

Lift status — OPEN, with a DERIVED-CONDITIONAL path. Granting the Kaloper–Padilla import (externally PROVEN) and the identification of the corpus Z_2-even four-form with the sequestering global constraint, R0 follows — **DERIVED-CONDITIONAL** on that identification, which is itself a finite formalization task (ZS-A28), not a new sector. The cost A27 attaches to channel (ii) — a new continuous sector (Exhaustion case c) — does **not** apply here: sequestering adds a rigid global Lagrange multiplier, with no β-function and no new particle. This is the single highest-value first move for B3 (§8.1): R0 is logically prior to M1, it has the easiest external-proven lift, and it incurs no new-sector cost.

§5. Root M1: the four-form residual programme

The retracted R3 manufactured a "freeze" by halting a toy integration; continued to late times the field relaxes to zero, and a constant Ohmic friction gives $\int\gamma\, dt = \infty$, so it cannot freeze (this also refutes the v1.3 Caldeira–Leggett repair). The clean replacement needs no friction and no freeze time: a four-form that stores the electroweak transition *topologically*.

Let $q_- = {*F_4^-}$ be a Z_2-odd four-form with the minimal action $S_- = \int\sqrt{-g}[-q_-^2/(2\chi_Z) + q_-\, O_{EW}^{(-)}/\bar{M}_P^2]$. Varying the three-form potential gives a conservation law solved by $q_- = \chi_Z(\mu_- + O_{EW}^{(-)}/\bar{M}_P^2)$. If exact Z_2 forces the integration constant $\mu_- = 0$, then after the transition

$$q_- = \chi_Z\, O_{EW}^{(-)}/\bar{M}_P^2 \ \ \text{(conserved)}, \quad \rho_\Lambda^{rem} = q_-^2/(2\chi_Z) = \chi_Z(O_{EW}^{(-)})^2 / (2\,\bar{M}_P^4) \ > 0.$$

Because a four-form has no local propagating mode, this residual is automatically positive, exactly constant, and free of any friction or cutoff (verified). Its relation to ZS-A27's integration-constant result is exact: that result says Λ is generically a free integration constant; M1's content is to use exact Z_2 to *fix* that constant ($\mu_- = 0$) for the odd mode.

Theorem 2 (Four-Form Electroweak Residual, M1). A Z_2-odd four-form linearly coupled to a dimension-4 electroweak order-parameter operator yields a positive, exactly constant vacuum residual $\rho_\Lambda = \chi_Z(O_{EW}^{(-)})^2/(2\bar{M}_P^4)$. **Status: HYPOTHESIS-strong**, conditional on the four results in §5.1.

§5.1 The four conditions M1 must still meet

(i) It must be a memory, not a tether. The conservation law makes q_- track the *final* value of $O_{EW}^{(-)}$; that is genuine *memory* of the past transition only if the transition produces a membrane jump, a residual topological label, or a fixed superselection sector. As written it is algebraic locking to the final order parameter; the membrane/label structure must be exhibited.

(ii) A nonzero Z_2-odd dimension-4 operator is the hardest condition. The standard electroweak invariant $H^\dagger H$ is Z_2-*even*, so the operator $O_{EW}^{(-)}$ entering the residual cannot be it. A nonzero Z_2-odd dimension-4 operator must come from a genuine odd structure — a difference of two Higgs copies, a sector-exchange-odd combination, an orientation-sensitive anomaly, or a boundary transgression term — and it must be *derived* from the master action, not posited. The corpus does not yet define this operator.

(iii) The integration constant cannot simply be declared zero. Forcing $\mu_- = 0$ requires showing that μ_- is an odd superselection label, that the physical boundary state is Z_2-invariant, and that neither spontaneous breaking nor flux tunnelling regenerates it. If $O_{EW}^{(-)} \neq 0$ arises from spontaneous Z_2 breaking, the vacuum structure must explain why μ_- stays protected while O_- does not.

(iv) Loop mixing is uncomputed. Quantum corrections may generate q_+q_-, q_-R, $q_-H^\dagger H$, or $(F_4^-)^2(F_4^+)^2$. Any one re-mixes the even sequestering sector with the odd residual, potentially re-importing the offset that sequestering removed. This has not been calculated.

Until (i)–(iv) are met, M1 is a well-motivated *ansatz*, not a result — and saying "M1 is the electroweak memory" is premature. The honest phrasing is that M1 is a *programme* with a positive structural core and four named obstructions.

§5.2 M1 as a topological residual, its external precedent, and the gap caveat

The motivating intuition — that the residual is a Yang–Mills-type *minimum excitation*, a topological mass gap rather than a continuously-relaxing field — is **a viable candidate analogy** for the microscopic scale entering M1, and it has direct external precedent. Urban and Zhitnitsky [3,4] derive a cosmological vacuum energy from the QCD *topological* sector (the Veneziano ghost of the $U(1)_A$ problem): $\rho_\Lambda \simeq H \cdot m_q\langle\bar{q}q\rangle/m_{\eta'} \sim (3.6\text{–}4.3\times10^{-3}\ \text{eV})^4$, strikingly close to observed. The key is that a *non-local, topological* vacuum energy survives to the infrared because the ghost's finite-volume properties deviate (as $\sim H/\Lambda_{QCD}$) from their Minkowski values. Klinkhamer–Volovik q-theory [5] is the four-form realization of the same idea and is the closest external analog of M1.

The gap caveat (an external-review correction). The Q7 master equation gives a dimensionless spectrum $\lambda(\lambda+2A/Q)(\lambda+A) = 0$ with roots $\{0, -2A/Q, -A\}$. These are eigenvalues of a *Liouvillian / Lindblad generator*, so the nonzero ones are **relaxation rates** — $\rho(t) - \rho_{ss} \sim e^{-\Delta t}$ — not Hamiltonian energy levels. A Yang–Mills mass gap is $E_1 - E_0$ (equivalently the decay of a Euclidean correlator). These are not the same quantity (*verify_v17.py*, LG1). So $\Delta_{\mathcal{L}} = A\Gamma_0$ is a **relaxation gap, not yet a physical mass gap**; promoting it requires an explicit map $L_{Q7} \to H_{Z,-}$ (or $\mathbb{T}_- = e^{-aH}$) with proven spectral equivalence.

The honest statement is that the *dimensionless* Liouvillian relaxation gap is exactly computable; its identification with a physical odd-sector mass scale is OPEN.

What actually enters B3: a topological susceptibility, not the relaxation gap. Following the Veneziano-ghost precedent, the quantity that feeds the vacuum energy is the *odd-sector topological susceptibility* $\chi_- = \int d^4x\, \langle \Omega_-^{(4)}(x)\, \Omega_-^{(4)}(0)\rangle_c$, which non-perturbatively scales as $\chi_- = c_\chi \Lambda_-^4$ for an odd-sector strong scale Λ_-. With a matching $O_{EW}^{(-)} = c_{match}(A,Q)\cdot\chi_-$, the residual takes the form

$$\rho_\Lambda^{pred} = C_-(A,Q) \cdot \Lambda_-^8 / \bar{M}_P^4, \qquad C_- = \chi_Z c_{match}^2 c_\chi^2/2 \ \ (\textit{verify\_v17.py}, \text{TS1}).$$

This relocates the open problem cleanly: B3 closes when Λ_-, c_χ, c_{match}, and χ_Z are all derived from the master action — not when a gap is reinterpreted as a mass. **The decoupling point, stated correctly:** a Wilsonian EFT decouples heavy *local* operators, but a topological four-form has *no local propagating degrees of freedom*, so conventional local-particle decoupling does *not by itself* exclude a topological M1 sector. It does not, however, prove M1 survives: M1's RG stability (operator mixing, condition (iv)) remains OPEN. The earlier "decoupling objection answered (MG5)" was an interpretation, not a computed check, and is no longer counted in the audit.

§5.3 The odd-operator multiplicity, and the selector-free reformulation

Condition (ii) of §5.1 asked for a *unique* $\mathbb{Z}_2$-odd dimension-4 operator. The corpus's only explicit J_Z-graded dimension-4 BRST cohomology — the Kostant cubic-Dirac complex of ZS-M27 (Theorem M27.2, **DERIVED**) — realizes the J_Z-odd ($\Gamma = -1$) subspace as **two-dimensional**, $H_D^- = \mathrm{span}\{\psi_{-3}, \psi_{-11}\}$: the two ramified-prime characters of $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ (ZS-M22). Their product $\chi_{-3}\cdot\chi_{-11} = \chi_{33}$ is J_Z-even, so no canonical single odd generator exists without breaking the 3↔11 symmetry. *This corrects an earlier internal reading* that took the odd cohomology to be empty: the seam-grading (even = 4, odd = 0) of ZS-M28 Theorem 28.12 is a *different, non-isomorphic* grading on the minimal cobordism slice, not the Clifford grading relevant here. Condition (ii) is therefore not a fill-in; it is a genuine **multiplicity-2**.

Two reformulations promise to remove the multiplicity *without* selecting a generator. They do not fare equally, and the difference is decided by one parity bookkeeping.

Reformulation A (determinant line) — parity-obstructed. The top exterior power $\Lambda^2 H_D^-$ is one-dimensional by elementary multilinear algebra, $\Psi_{det} = \psi_{-3} \wedge \psi_{-11}$, and any basis change $U \in U(2)$ acts on it only by $\det(U)$. It is genuinely canonical — but it is the **wrong parity**. Since $\gamma^5 = J$ (ZS-S15), J_Z acts as -1 on *each* odd character, hence on Ψ_{det} by $(-1)(-1) = +1$: Ψ_{det} is J_Z-**even**. M1 requires a J_Z-**odd** four-form ($H^{0,4}_{J=-}$). The exchange involution $C_{3\leftrightarrow 11} : \psi_{-3} \leftrightarrow \psi_{-11}$ that renders Ψ_{det} odd is a *different* $\mathbb{Z}_2$ from the seam J_Z (C has eigenvalues ±1; $J_Z = -1$ on the entire odd plane), so "C-odd" is not "J_Z-odd." In general the J_Z-parity of a k-linear in odd fields is $(-1)^k$, so the *even*-degree objects — the determinant and every bilinear — are J_Z-even, and no wedge or determinant can be the J_Z-odd operator M1 needs. (Odd-*degree* covariants do exist in general, e.g. a cubic $(\Omega_-^\dagger K_-\Omega_-)\Omega_-$; but in the 2-dim odd algebra $\Lambda^3 H_D^- = 0$, so the determinant-line construction leaves only the linear multiplet itself.) A parity-preserving transgression cannot carry the J_Z-even Ψ_{det} into the J_Z-odd cohomology. Status: **NON-CLAIM** (parity-obstructed), not merely OPEN.

Reformulation B (cross-coupled quadratic invariant) — the viable reframing. Do not seek a single four-form; carry the J_Z-odd doublet $\Omega_- = (\Omega_{-3}, \Omega_{-11})^T$ and let the residual be the quadratic invariant

$$\rho_\Lambda = (1 / 2\bar{M}_P^4)\ \Omega_-^T K_- \Omega_- .$$

This is J_Z-**even** (quadratic in odd fields, k = 2) — correct, since the vacuum-energy density is a parity-even scalar — and it is basis-independent *provided K_- transforms as the induced operator on H_D^-*

($\Omega_- \to U\Omega_-$, $K_- \to UK_-U^\dagger$); it is not invariant if Ω_- alone is rotated with K_- held fixed. With K_- covariant, **no generator selection is required**. The over-strong reading "multiplicity-2 cannot be closed" is thereby *refuted*: the multiplicity obstructs only a spurious unique generator, never the physical scalar.

The kernel is not free. The corpus forbids the naive block-diagonal V_4 sum: ZS-M31 Lemma M31.0 with ZS-M33 Theorem M33.1 (**PROVEN**) show the V_4 regular representation collapses under Schur orthogonality to a block-diagonal sum-form that violates Non-Separability, mirroring the PROVEN block-Laplacian $L_{XY} \equiv 0$ and the Cross-Coupling Theorem (ZS-M2, **PROVEN**: every force formula involves all three sectors). The corpus-admissible object is the Z-mediated, non-separable Path-γ joint operator of ZS-M33 (Reading C, Theorem M33.3, **DERIVED**): X(g) acting on $H_{BFV} \otimes H_{arith}$, sandwiched by the Z-mediator projector $\Pi_Z = ½(I + J_Z)$, the Wilson-LOCATOR prime-phase forcing non-separability. The odd effective kernel is then $K_- = P_- X^\dagger X P_-$ (P_- the projection onto H_D^-), positive-semidefinite by construction.

The decisive, computable falsification. K_- is a 2×2 PSD matrix, and its rank classifies the outcome. **Case A — rank 1** ($\det K_- = 0$): the cross-coupling dynamically selects a unique odd direction $|\Omega_{phys}\rangle$ (the nonzero eigenvector), and the selector problem closes with *no* external choice. **Case B — rank 2** ($\det K_- > 0$): there is no unique linear generator, but ρ_Λ is the unique quadratic invariant, so closure is selector-free. **Case C — rank 0** ($K_- = 0$): the odd arithmetic sector does not couple to the residual and this route terminates. Cases A and B both close the multiplicity; only C kills it. The rank is a basis-independent CP-invariant: $K_- = P X^\dagger X P$ is a Gram matrix, so rank K_- = rank X by elementary linear algebra, hence basis-independent. *The further reading* "rank K_- = the Kraus rank of an A27 vacuum CPTP channel" would require an explicit channel whose Kraus operators are the columns of X — and ZS-M33's X(g) is an RH/Weil joint operator, not (yet) that channel — so the **Choi–Stinespring / Kraus interpretation is OPEN, not DERIVED** (an over-claim corrected from v2.0). What is settled is only the basis-independence of the rank.

What this settles, and what it does not. The reformulation *recasts* condition (ii) / gate B3-2 from "prove $\dim H^{0,4}_{J=-} = 1$" (false in the M27 realization) to "compute rank K_-" — a finite, decidable test (DERIVED that the reframing is well-posed). It does **not** close B3-2: $K_- = P_- X^\dagger X P_-$ uses ZS-M33's X(g), which is constructed and proven non-separable in the RH-bridge / Weil-positivity setting for ζ_K, not in the vacuum-energy setting of this paper; ZS-M33 itself limits the correspondence to a structural isomorphism, not a physical identity. Importing X(g) into the four-form residual is a **structural transplant — HYPOTHESIS**, and the rank of K_- in the A27 context is **uncomputed (TARGET-COMPUTATION)**. The other roots are untouched: R0 ($\mu_- = 0$ / even-sequestering) stays OPEN, the physical scale Λ_- (S1) stays OPEN, and the blind ρ_Λ prediction remains. B3 therefore remains OPEN; multiplicity-2 is reframed, not closed.

The geometric legs (now DERIVED), and the one back-fit to avoid. The geometric asymmetry that motivates the doublet's two "legs" is corpus-grounded, and is stronger than this section's first draft recorded: χ_{-3} sits on the Y-sector pre-truncation icosahedron (ZS-M28 Theorem 28.14) and χ_{-11} on the Q = 11 register (ZS-M22). **ZS-M30 Theorem 30.3** (Robin-Truncation Boundary Family, **DERIVED**) locates the χ_{-3} Eisenstein carrier as the t = 0 endpoint of an action-derived Robin family $h(x; t) = A\cdot K(x; t)$ — from the non-minimal coupling $F(\Phi) = 1 + A|\Phi|^2$ (ZS-F0 §8.2 Theorem 8.1, PROVEN) — explicitly promoting Theorem 28.14 from DERIVED-CANDIDATE to **DERIVED**. The two legs are therefore graded **DERIVED** (χ_{-3}, via M30) and **PROVEN** (χ_{-11}, via M22), which makes the cross-coupled doublet geometrically natural — two distinct, derived homes are more consistent with cross-coupling than with selecting a "winner." This strengthens the doublet but does **not** by itself close B3-2: the dynamical kernel transplant (the ZS-M33 import) and its rank remain. The one genuine back-fit to avoid is the conductor ordering log 3 < log 11 as a selector: ZS-M28 shows that inserting the conductor decoration diag(0, log 3, log 11, log 33) into the Kostant Dirac *breaks* the chirality grading or annihilates the four-dimensional cohomology, so a "smaller-conductor ground state" conflicts with the corpus.

Multiplicity-2 reformulation status (§5.3) — new gate labels carried to ZS-A28.

Result	Statement	Status
Multiplicity-2 of cond. (ii)	$H_D^- = \mathrm{span}\{\psi_{-3}, \psi_{-11}\}$ is 2-dim; the product χ_{33} is even (ZS-M27).	**DERIVED**
Determinant-line path	$\Psi_{det} = \psi_{-3} \wedge \psi_{-11}$ is J_Z-even; cannot source a J_Z-odd 4-form.	**NON-CLAIM** (parity)
Selector-free invariant	$\rho_\Lambda = \Omega_-^\dagger K_- \Omega_-$ basis-independent (K_- covariant); rank basis-independent (Kraus/CPTP reading OPEN).	**DERIVED**
Corpus kernel K_-	$K_- = P_- X^\dagger X P_-$ via ZS-M33 Path-γ X(g); transplanted to A27.	**HYPOTHESIS**
Rank test (B3-2)	rank $K_- \in \{1$: unique dir.; 2: selector-free; 0: decouple$\}$.	**TARGET-COMP.**
Geometric two legs	χ_{-3}↔Y-icosahedron (ZS-M30 Thm 30.2/30.3, promotes 28.14); χ_{-11}↔Q-register (ZS-M22).	**DERIVED** / **PROVEN**
B3-2 / B3 overall	Reframed to a finite rank test; transplant + rank + μ_-=0 remain.	**OPEN**

§6. Root S1: the blind scale, and what the mass gap does to it

With Klinkhamer–Volovik scaling $\rho_\Lambda \sim E_*^8/\bar{M}_P^4$ (equivalently $\rho_\Lambda^{1/4} = E_*^2/\bar{M}_P$), the observed meV density needs $E_* \approx 2.335$ TeV. The electroweak vev alone, v = 245.93 GeV, gives $\rho_\Lambda^{1/4} \approx 2.5\times10^{-5}$ eV (verified) — about 90× short, ~6×10^7 in density. This missing factor is **S1**, the third root. The mass-gap intuition is the natural way to try to supply it; here is exactly what it does and does not do.

§6.1 What the mass gap does NOT license: E

A tempting closure sets $E_* = (Q - \dim Z)\cdot v = 9v \approx 2.21$ TeV, replacing the author's rejected $3\pi v \approx 2.32$ TeV. **This is the same numerology with a different integer**. The required factor is $E_*/v \approx 9.50$, and no clean corpus constant supplies it ($3\pi \approx 9.42$, $Q - 2 = 9$, $\sqrt{(Q/A)} \approx 11.7$, $1/A \approx 12.5$; nearest is 3π, off by 0.07 — verified). More fundamentally, a genuine mass gap is *not* an integer multiple of an input scale: in Yang–Mills it is dimensional transmutation, $\Lambda \sim \mu\cdot\exp(-1/2b_0g^2)$, a transcendental function of the coupling that spans orders of magnitude (verified, factor ~2000 across g = 0.3–2.0). So $E_* = 9v$ is doubly barred — post-hoc, and structurally inconsistent with the gap mechanism it invokes. **9v and 3πv: REJECTED-COINCIDENCE.**

§6.2 What the relaxation gap does and does not supply

The Q7 spectrum $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$ (ZS-Q7, DERIVED) gives nonzero *dimensionless* rates $2A/Q \approx 0.015$ and $A \approx 0.080$. As §5.2 established, these are eigenvalues of a *Liouvillian* generator — **relaxation rates**, not a Hamiltonian energy gap — stated in units of an attempt scale Γ_0. So the corpus supplies an exact *dimensionless relaxation gap* and nothing dimensionful: no physical odd-sector mass scale follows until either an explicit correspondence $L_{Q7} \to H_{Z,-}$ (or $\mathbb{T}_- = e^{-aH}$) is constructed, or an independently derived confining sector fixes Λ_-. The earlier wording — "the gap is finite-dimensional and exactly computable; the dimensionful gap is $A\cdot\Gamma_0$" — conflated the two and is withdrawn: $A\cdot\Gamma_0$ is a dimensionless rate times an undetermined attempt scale, not a derived mass. This is the same single missing dimensionful input that §7 shows is forced by dimensional analysis.

§6.3 The fork: "2.3 TeV" is an artifact of the scaling choice

The TeV requirement is specific to the Klinkhamer–Volovik scaling $\rho \sim E_*^8/\bar{M}_P^4$. The Zhitnitsky scaling is different: $\rho \sim \Lambda^3\cdot H$, with one power of the infrared Hubble scale. Taking $\Lambda \sim \Lambda_{QCD} \approx 0.1$ GeV and $H \approx 10^{-33}$ eV gives $\rho_\Lambda^{1/4} \approx 6\times10^{-3}$ eV (verified) — the observed order of magnitude with **no TeV input at all**. The price is twofold: the residual then *tracks H*, i.e. it is dynamical dark energy rather than a constant Λ, and it requires the $U(1)_A$-ghost structure. So the question "what scale does M1 need?" is not even well-posed until the residual's *scaling* is derived from the master action. Under one scaling S1 demands a 2.3 TeV gap; under the other S1 dissolves but a time-dependent residual and a ghost sector appear. **S1 is therefore reframed**: not "an arbitrary missing 2.3 TeV," but "derive the residual's scaling (KV vs Zhitnitsky) and its dimensionful gap (Γ_0) from the action." **Status: OPEN**, now with a definite shape.

§6.4 The dark-energy lever, and why it does not close B3

v1.7 tried to turn the dark-energy equation of state into a closure lever; external review showed that attempt over-reached, and it is corrected here. The structural observation survives: the two scalings of §6.3 correspond to two physical realizations of a residual — a *constant* ($\mu_- = 0$; $w = -1$) or an *H-tracking* one (w evolving). But the three claims v1.7 layered on top do not.

(1) $\Lambda_Z \approx 25.6$ MeV was back-solved, not predicted. It was computed as $\Lambda_Z = (\rho_{\Lambda,obs}/H_0)^{1/3}$ — its inputs *are* the observed dark-energy density and the Hubble scale (*verify_v18.py*, RET-ZH). That is an inversion of the target, logically identical to back-solving the 2.335 TeV of §6.1; only the scaling moved the target from TeV to MeV. **Withdrawn:** "natural IR-scale prediction," "near-closure," "no scale was back-solved." Correct status: Λ_Z is a *diagnostic* value — the effective ghost scale that the observed numbers require.

(2) The two scalings cannot be mixed. The dynamical form $\rho \propto \Lambda^3H$ and the constant form $\rho = C_-\Lambda^8/\bar{M}_P^4$ are not the same formula for one Λ. Substituting the back-solved $\Lambda_Z = 25.6$ MeV into the constant form gives $\rho_\Lambda^{1/4} \approx 2.7\times10^{-13}$ eV — about 8×10^9 times too small (*verify_v18.py*, RET-MIX). So the v1.7 chain "$\Lambda_Z \approx 26$ MeV $\Rightarrow \chi_- = c_\chi\Lambda_Z^4 \Rightarrow \rho_\Lambda = C_-\Lambda_Z^8/\bar{M}_P^4$" does not hold. **The scaling must be derived from a common action first; w(z) does not pick it.** Until an action $S_{common}[q_-, g, \Omega_-]$ is exhibited whose distinct saddles give $\rho_{const} \sim \Lambda^8/\bar{M}_P^4$ and $\rho_{dyn} \sim \Lambda^3H$, these are *two competing phenomenological scalings*, not two branches of one M1.

(3) The ghost w_0 is a toy discriminator, not a confirmed prediction. Solving $\rho_{DE} = \alpha H$ with the Friedmann equation gives $w_0 = -1 + (1 - \Omega_{DE})/(2 - \Omega_{DE}) = -0.763$ for $\Omega_{DE} \approx 0.69$ — but this is a structural consequence of the *assumed* scaling $\rho_{DE} = \alpha H$, not a derivation of it from the master action. Three caveats make the "0.9σ agreement with DESI DR2" non-evidential (*verify_v18.py*, RET-DE): a single coordinate w_0 is not the correlated (w_0, w_a) likelihood; the ghost predicts $w_a > 0$ while DESI prefers $w_a < 0$ (phantom crossing); and the linear ghost has a known $c_s^2 < 0$ instability [12]. A real test needs $\Delta\chi^2$ / AIC / BIC / Bayes against DESI BAO + CMB + SNe, plus $c_s^2 \geq 0$, $Q_s > 0$, and growth.

What survives. The dark-energy data remain a *future falsifier* of a completed branch — not an input that supplies one. DESI DR2's 2.8–4.2σ preference for evolving dark energy [9,10] makes the dynamical realization worth pursuing in ZS-A28, and the ghost $w_0 \approx -0.76$ is a useful discriminator to keep in view.

But the choice of scaling, and the one missing scale, must come from the action; the deeper reason is the dimensional no-go of §7.

§6.5 The scale route: dimensional transmutation, not an integer multiple

Whichever scaling is eventually derived, S1 needs exactly one dimensionful scale Λ_-. The right construction (channel (ii) of §7, the spectral-action route) is not an integer times v but **dimensional transmutation** of a J_Z-odd, asymptotically-free factor G_- whose coupling is fixed by the geometry,

$$\Lambda_- = \mu_0 \exp[-\, 8\pi^2 / b_0 g_-^2(\mu_0)] \cdot (\ldots), \quad g_-(\mu_0) = f(A, Q),$$

with μ_0 an already-defined reference ($\bar{M}_P$, v, or a corpus threshold — not a new fitted scale). A single coupling spans MeV-to-TeV scales as a smooth function of g_- (*verify_v18.py*, DT1/DC2), so Λ_- would be an output of a beta function rather than a hidden choice. But §7 shows this channel imports a NEW continuous sector (a finite Z-register has no beta function), so it is Exhaustion case (c), outside the locked corpus. **Immediate-rejection conditions:** no asymptotically-free odd factor; g_- a free parameter; μ_0 a new fitted scale; or Λ_- selected after seeing the observed value — any one of these and S1 does not close.

§7. Scale generation: the A–Q-only no-go, and the open routes

Across A22–A27 the absolute scale has resisted every tool. There is a single structural reason, and stating it is the main result of this paper. **The locked dimensionless data cannot fix a dimensionful scale on their own.** Every locked invariant — A = 35/437, Q = 11, (Z,X,Y) = (2,3,6), the i-tetration multiplier $|\mu|$ = 0.89151, the Q7 relaxation rates {0, 2A/Q, A}, the embedding weights (3,2,6)/11 — is a pure number. The cosmological constant is *dimensionful*.

Theorem (A–Q-Only Scale-Generation No-Go, DERIVED). A dimensionful quantity cannot be a function of dimensionless inputs alone (Buckingham-π). Therefore A, Q, and the dimensionless finite-register invariants *on their own* cannot define the dimensionful ρ_Λ; closing B3 requires at least one dimensionful normalization or a dynamical scale-generating mechanism. This is almost trivial — and it is the honest content of what v2.0 called the "Dimensional Closure Theorem," which over-stated it. *Two corrections (external review).* First, the full corpus is *not* "purely dimensionless": the one locked dimensionful object $\bar{M}_P$ (and, downstream, v and Λ_{QCD}) are dimensional, so $\rho_\Lambda = \bar{M}_P^4 \cdot f(A,Q,\ldots)$ is dimensionally admissible — the difficulty is the **hierarchy** $f \approx 10^{-121}$ ($\rho_\Lambda/\bar{M}_P^4 \approx 7\times10^{-121}$, *verify_v18.py*, DC0), a selector problem, not a dimensional one. Second, Buckingham-π fixes the dimensional *form* but says nothing about the *magnitude* of f; the routes below are therefore a **non-exhaustive taxonomy of currently-known mechanisms (HYPOTHESIS)**, not a proven trichotomy — full action extensions, the SM running sector, gravity anomalies, topological quantization, and nonperturbative saddles are not excluded:

Route (known, non-exhaustive)	Mechanism	Verdict
(i) Target-fit exponent	read an exponent off the answer: $A^n = 10^{-121}$ needs n = 109.6; $(2A/Q)^n$ needs n = 65.4 (verify_v18, DC1)	**NUMEROLOGY** — barred (the form of 9v, 3πv, π/A)
(ii) Action-fixed transmutation	$\Lambda_- = \bar{M}_P \exp(-8\pi^2/b_0 g_-^2)$ with b_0, g_- *fixed by the spectral action*, reaching MeV–TeV from $\bar{M}_P$ (DC2)	**LEGITIMATE** nonperturbative physics *if* g_- is corpus-forced (a live

Route (known, non-exhaustive)	Mechanism	Verdict
		path); a new continuous sector only if it is not
(iii) Observed Hubble scale	$\rho \sim \Lambda^3 H$; smallness = $(\Lambda/\bar{M}_P)^3(H/\bar{M}_P)$, $H/\bar{M}_P \approx 6\times10^{-61}$ (DC3) — no tuned exponent	**DYNAMICAL DE** — closes B3 relative to a measured input; ρ_Λ then time-dependent ($w \neq -1$)

Corollary. The corpus can supply the *dimensionless* coefficient $C_-(A,Q)$ and the M1 coupling *structure*; from A and Q *alone* it does not supply the scale. But it *can* generate the scale dynamically from the locked $\bar{M}_P$ via route (ii), provided the odd coupling $g-$ is forced by the existing product algebra rather than fit. This makes "close B3" precise: *commit to (ii) or (iii) and fix that scale from real dynamics — an action/topology-fixed exponent, not an inverted target.* It also explains the failures honestly: 9v, 3πv, π/A and the back-solved 2.3 TeV / 25.6 MeV were target inversions (route i, or circular), not action-fixed derivations — a category distinct from a legitimate transmutation, which v2.0 wrongly conflated with the barred case.

Relation to A26 and §2.1. A26 showed the tools are *relative* (blind to a central shift); §2.1 showed Λ is an *integration constant* in emergent gravity. The A–Q-only no-go is the third face of the same fact: the dimensionless data alone do not fix the integration constant's *value* — though, as §7.1 computes, the locked $\bar{M}_P$ with an action-fixed transmutation can supply a scale. **Status of the no-go: DERIVED. Status of B3 itself: OPEN** — now with its openness explained and the known routes classified.

§7.1 Channel (ii), computed: the transmutation frame — validated, and bounded by an open spectral invariant

§7 leaves route (ii) — an action-fixed dimensional transmutation from the locked $\bar{M}_P$ — as the one no-observation path to the scale. We compute it here, blind, on the corpus's own machinery, and report both what it settles and what it does not. The anti-numerology discipline is pre-registered: **no epoch** (no ν_{now}), no observed ρ_Λ as a derived input, and no spectral invariant fit to the answer.

The machinery is DERIVED from PROVEN spectral inputs. Within the corpus the coexact spectrum is PROVEN, while the UV prefactor and the determinant relation are DERIVED; the Higgs VEV then follows by a dimensional transmutation — the Factorized Determinant Theorem (ZS-S1 §6.12), itself DERIVED:

$$v = \bar{M}_P \cdot \exp[-\gamma_{CW} \cdot C_M^{sp}] = \bar{M}_P \cdot \exp[-(38/9)(11 \ln 2 + \ln 3)] = \bar{M}_P \cdot \exp(-36.83) = 245.93\ \text{GeV},$$

where $\gamma_{CW} = 38/9$ is the Coleman–Weinberg coefficient and $C_M^{sp} = 11 \ln 2 + \ln 3$ the even-sector spectral log-determinant. We ride this proven machinery, not a new one.

Applied blind to B3. Under the eighth-power (Klinkhamer–Volovik) scaling $\rho_\Lambda = E_*^8/\bar{M}_P^4$, the only scale the proven transmutation hands over is $E_* = v$ itself, so

$$\rho_\Lambda^{1/4}|_{blind} = E_*^2/\bar{M}_P = v^2/\bar{M}_P = 2.48\times10^{-5}\ \text{eV}.$$

This uses no epoch and no observed input — it is the genuine blind output, and it reproduces A27 §6's documented S1 figures (required $E_* = 2.335$ TeV, 90× short).

The blind benchmark, then a post-unsealing comparison. The output above is blind. Only now do we compare it to the observed value; set against the additive cutoff^4 catastrophe, the three views of the 120-order problem are:

View of B3	$\rho_\Lambda^{1/4}$	density miss
X-sector (continuum, cutoff^4)	$\sim \bar{M}_P \sim 2.4\times10^{27}$ eV	~120 orders
Y-sector (BLIND transmutation, §7.1)	2.48×10^{-5} eV	~8 orders
observed	2.24×10^{-3} eV	—

On this *post-unsealing* comparison the transmutation lands **within ~8 of the 120 orders** (a ~112-order improvement over the additive cutoff^4 estimate). This is *suggestive* that the category reframing (a finite, topological vacuum source, not an additive cutoff^4) points in the right direction. It does **not** by itself establish that M1 carries the eighth-power (KV) scaling, that the odd topological susceptibility shares the Higgs exponent, that the four-form is normalized by v^4, that R0 leaves precisely this residual, or that the matching coefficient is unity — each is a separate open computation. The honest statement: the existing electroweak hierarchy $v/\bar{M}_P \approx 10^{-16}$, inserted into the *assumed* eighth-power scaling, yields a small blind benchmark; the M1 mechanism itself remains unvalidated. The output 2.48×10^{-5} eV is blind; the figure "112 of 120" is a comparison defined only after seeing $\rho_{\Lambda,\mathrm{obs}}$.

But it does not close — and the gap is the documented trap. The pre-registered verdict is a MISS: 90× short. The residual ×9.5 = E_*/v is exactly the S1 gap of §6: the nearest clean corpus constant is $3\pi = 9.42$ (off by 0.07, already rejected as numerology; $9 = Q-2$ also rejected). Equivalently the target exponent $8\cdot A_{\mathrm{comp}} = 276.64$ does *not* arise from the blind even-sector geometry (294.65); it is reached only via $2\nu_{\mathrm{now}}\pi/A$ with $\nu_{\mathrm{now}} = 3.526$ (a forbidden epoch input) or $32\pi^2/(b_Z A)$ with $b_Z = 14.25$ (an underived BRST coefficient). The gap is the documented trap, not a closure.

Why even-sector, and what closure needs. The invariant used is the **even** (Higgs) sector's C_M^{sp} — the only spectral data the corpus currently hands over. But M1's four-form is **Z_2-odd**. Real closure therefore requires the **odd**-sector spectral log-determinant, which must *force* the exponent $A_{\mathrm{comp}}^{\mathrm{odd}} = 36.83 - \ln(9.5) = 34.58$ — i.e. $C_{\mathrm{odd}}^{\mathrm{sp}} = 8.190$, about 6% below the even $C_M^{\mathrm{sp}} = 8.723$ — from the actual odd eigenvalues, not by fitting. The value 8.190 is *itself back-solved* from the observed ρ_Λ (via $E_*^{\mathrm{need}} = \sqrt{(\rho_{\mathrm{obs}}\bar{M}_P)} = 2.335$ TeV): it is a pre-registered **falsification target** for A28, **not** a prediction, closure theorem, or blind output — and A28 must compute the odd spectrum without using it.

Anti-numerology: the natural odd invariants bracket but miss. The natural odd invariants — small-integer combinations of the corpus building blocks {ln 2 (the Z ln 2-capacity, ZS-A22), ln 3, ln 11 (the ramified conductors)} — *bracket* the observed value but do not hit it:

odd spectral invariant	$\rho_\Lambda^{1/4}$	vs observed
$10 \ln 2 + \ln 3 = 8.030$	8.7×10^{-3} eV	×3.9 over
(needed) $C_{\mathrm{odd}}^{\mathrm{sp}} = 8.190$	2.24×10^{-3} eV	observed
$12 \ln 2 = 8.318$	7.5×10^{-4} eV	×3 under

The target $C_{\mathrm{odd}} = 8.190$ lies between them, reached only by a non-integer coefficient (e.g. $11 \ln 2 + ½ \ln 3$). A non-integer weight is *not* automatically numerology — half-integers arise legitimately from Pfaffians, reality / Gaussian-determinant factors, ghost subtraction, or representation multiplicity; the real test is whether the weight is fixed *a priori* by the operator spectrum and measure (legitimate) or chosen to hit the target (numerology). No spectrum-derived weight is in hand here, so a fitted ½ would be numerology. C_{odd} must be pinned to ~3% of the natural range, and the result is exponentially sensitive (a factor-1.6 coupling ambiguity moves $\rho_\Lambda^{1/4}$ across orders of magnitude). We do not have the odd eigenvalues, and we do not fabricate them.

Cross-check — the gauge route is the same exponent. Parametrizing the transmutation instead as an asymptotically-free odd confining sector, $\Lambda_{-} = \bar{M}_P \exp[-8\pi^2/(b_0 g_-^2)]$, the corpus polyhedral rule $b_0 = (V+F)/G$ (ZS-S1) fixes the β-coefficient, and the QCD sanity check ($b_0 = a_3 = 23/3$, $\alpha_s = 11/93$) reproduces $\Lambda_{QCD} \approx 89$ MeV. But with the Z-sector's own b_0 (the tetrahedron family, $(V+F)/G \in \{4/3, 5/3\}$) and the spectral-action coupling $g_-^2(\bar{M}_P) \approx 0.3$–$0.5$, the transmutation gives $\Lambda_{-} \approx 10^{-64}$–$10^{-23}$ GeV across that range — tens of orders (≈40–80) below the Planck scale (the verification's representative point $b_0 = 5/3$, $g_-^2 = 0.4$ gives 8.9×10^{-34} GeV, 51 orders below): the Z-sector does *not* confine, reproducing ZS-A22's barrier B2 (the Z-sector is a rank-≤2 mediator, not a propagating gauge field) by direct computation. The eighth-power target $\Lambda_{-} \approx 2.3$ TeV requires $b_0 g_-^2 = 2.283 = 8\pi^2/34.58$ — identical to the Higgs route's exponent. The two parametrizations are **one exponent**; the Higgs / Factorized-Determinant route is the better-grounded one (PROVEN to give v, 90× short rather than 47 orders off), and both isolate the odd spectral invariant as the single missing quantity.
Status. Channel (ii): the **blind KV-scaling benchmark is computed** (2.48×10^{-5} eV); the 112/120-orders figure is a post-unsealing comparison and the M1 mechanism is unvalidated, so the frame is **HYPOTHESIS** (not "validated"). **B3 closure via (ii) is OPEN** — blind gives 2.48×10^{-5} eV (90× short), the closing ×9.5 is the documented trap, the back-solved target $C_{odd} = 8.190$ is uncomputed from first principles, and — critically — even a correct odd invariant does **not** by itself close B3 (§8.1, §8.2). Identifying that invariant with a Ray–Singer torsion is itself a **HYPOTHESIS**, not DERIVED-CONDITIONAL (§8.1): see the Odd Determinant–Torsion Identification Theorem and its even-dimension vanishing concern.

§8. What an absolute no-go would require

To upgrade B3 from OPEN to an *absolute* no-go — the claim that *no* admissible Z-Spin extension can produce a small positive residual — one would need a classification theorem, not another failed attempt:
Vacuum-Residual Exhaustion Theorem (target). For every diffeomorphism-invariant, gauge-invariant, radiatively stable, zero-free-parameter global extension of the Z-Spin master action, exactly one holds: (a) the vacuum residual is identically zero; (b) it depends on a free integration constant; (c) it depends on a new dimensionful input; or (d) if nonzero it breaks radiative stability or observational consistency.
ZS-A27 has not classified all admissible extensions — indeed it explicitly *admits* the four-form extension M1. The premise of an exhaustion theorem is therefore unmet, and any absolute-no-go claim would be premature. This is the structural reason the v1.3 "certified-unclosable" verdict was withdrawn, and why this version holds B3 at OPEN.

§8.1 The complementary question: what a closure would require

The exhaustion theorem above is the path to a *negative* verdict; external review supplied the matching *positive* checklist. Closing B3 by the four-root route {C1′, R0, M1, S1} — the *sequestered odd-sector topological-susceptibility* programme, $S_{close} = S_{ZS} + S_{seq}^{(+)} + S_{top}^{(-)}$ — means passing all seven of the following gates, in order. None is yet passed.

Gate	Required result
B3-1	Even sequestering removes the matter / EW / QCD / graviton constant shifts (root R0).

Gate	Required result
B3-2	Derive the physical odd multiplet Ω_-, construct the A27-specific positive kernel $K_-^{vac} = P_- X_{vac}^\dagger X_{vac} P_-$, and prove rank $K_-^{vac} \geq 1$ with basis-covariant matching (the §5.3 reframing; supersedes the former "dim = 1" requirement).
B3-3	A J_Z boundary theorem fixes $\mu_- = 0$ and the flux branch (memory as a flux jump, not a tether).
B3-4	An asymptotically-free odd factor G_- with coupling $g_-(\mu_0) = f(A,Q)$ derived, not fitted.
B3-5	Non-perturbative Λ_-, χ_-, and the matching coefficient computed (lattice permitted if the action is pre-fixed).
B3-6	Even–odd mixing and discrete anomaly exactly zero: $\gamma_{+-} = 0$ (else the offset re-enters).
B3-7	The blind $\rho_\Lambda^{pred} = C_-(A,Q)\Lambda_-^8/\bar{M}_P^4$ matches observation and passes perturbations / CMB / BAO.

Blind-prediction discipline. For any closure claim, the admissible inputs are A, Q, $\bar{M}_P$, and SM couplings already fixed independently; the forbidden inputs are H_0, $\rho_{\Lambda,obs}$, T_4^{now}, and "2.335 TeV." The gauge algebra, beta functions, $g_-(\mu_0)$, matching formula, branch rule, and counterterm basis are published and hashed *first*; ρ_Λ^{pred} is output exactly once and only then compared. If any gate fails, B3 stays OPEN or the route is RETRACTED — the same discipline that retracted C1, R3, R5, and A27.13.

ZS-A28: the next computations, in dependency order (corrected in v2.3 — the channel-(ii) invariant is one gate, not the gate).

The odd-invariant computation, in context. §7.1 isolates the odd spectral invariant as **one decisive S1 subgate** — not the sole remaining B3 computation. The blind benchmark is computed; closure does **not** hinge on a single number. The forward program still requires all of B3-1…B3-7 above. Within S1, A28 should compute the odd-sector invariant and test whether it *forces* the exponent $A_{comp}^{odd} = 34.58$ (the back-solved target $C_{odd}^{sp} = 8.190$ must **not** be used as an input). Even if it does, ρ_Λ is still not fixed: the coefficient $C_-(A,Q) = \chi_Z c_{match}^2 c_\chi^2/2$ and the eighth-power scaling form remain uncomputed.

Odd Determinant–Torsion Identification Theorem — OPEN (the torsion route is HYPOTHESIS, not DERIVED-CONDITIONAL). v2.2 identified the odd log-determinant with a Ray–Singer analytic torsion and called Cheeger–Müller a license to compute it. That identification is unproven, for three reasons. **(a)** A single coexact determinant is *not* the Ray–Singer torsion, which is the weighted alternating product over **all** degrees, $\log T_{RS} = \frac{1}{2}\,\Sigma_q(-1)^q q \log \det'\Delta_q^-$; reducing it to one coexact determinant needs a separate cancellation theorem. **(b)** The odd cohomology is non-acyclic ($\dim H_D^- = 2$, ZS-M27), so the torsion is not a single positive number but a *metric on a determinant line* (Poincaré–Reidemeister / Farber–Turaev), requiring a cohomology basis, determinant-line normalization, and explicit zero-mode handling. **(c)** Most sharply: Ray–Singer analytic torsion is **trivial (= 1, log = 0)** for closed orientable **even-dimensional** manifolds with unitary / orthogonal coefficients (Ray–Singer 1971; Cheeger–Müller [14] is stated for *odd*-dimensional manifolds). The Y-sector is **6-dimensional (even)**, so a unitary even-dimensional odd complex would give torsion ≡ 1 — exponent 0, *no* transmutation — **unless** the non-unitary arithmetic flat-bundle data (the conductors χ_{-3}, χ_{-11}) evades the vanishing. Whether it does is an open question requiring **external verification by a spectral-geometry / NCG specialist**. Status: **OPEN**; the analytic-torsion route is **HYPOTHESIS**, contingent on resolving (a)–(c).

Pre-registered decision rule (corrected). (i) *Blind*: no epoch (no ν_{now}), no observed ρ_Λ, no "2.335 TeV"; the invariant must come from the eigenvalues, not be fit to 8.190; registered before unsealing. **(ii) The matching exponent is NECESSARY, not SUFFICIENT.** Even if the odd invariant forces 34.58, B3 closes only when **all** gates close — R0 (the trunk, §4, logically prior), the remaining M1 conditions ($\mu_- = 0$, a derived $O_{EW}^{(-)}$, zero even–odd loop mixing), the coefficient $C_-(A,Q)$, the eighth-power form, and the

Selective Sequestering Compatibility of §8.2. The exponent alone does **not** close B3; v2.2's "wall = one number" framing is **withdrawn**. (iii) If the invariant gives a materially different exponent, or vanishes (the even-dimension concern), the torsion route is **demarcated / closed** and the scale is sought through R0 + channel (iii). Every outcome is a real result.
Order, corrected. The **top** priority is *not* the torsion number but the **Selective Sequestering Compatibility Theorem (§8.2)** and R0 (the trunk, §4). Without an action-level reason that sequestering removes the large constant vacuum energy *while leaving* the M1 residual, even a correct odd torsion does not close B3: the residual could be sequestered too, an integration constant could reappear, or even–odd mixing could reintroduce radiative instability. Attack §8.2 and R0 first; the odd-invariant computation is *one* gate to pursue in parallel, not *the* gate. The A18–A27 line diverged precisely by hitting the M1 / scale terminals before the trunk.

§8.2 The top open problem: the Selective Sequestering Compatibility Theorem

The single most important open problem for B3 is **not** the odd invariant but a compatibility theorem binding R0 and M1 in **one** action. Write the unified action $S_{unified} = S_{ZS} + S_{seq}^{(+)} + S_{odd}^{(-)} + S_{membrane}$, with $S_{seq}^{(+)}$ the Kaloper–Padilla global sequestering sector (§4), $S_{odd}^{(-)}$ the Z_2-odd four-form residual (§5), and $S_{membrane}$ the four-form flux / membrane sector. The theorem to prove is **selective** sequestering:

$$\partial \Delta V_i/\partial G_{\mu\nu}^{phys} = 0 \quad \text{(large even / matter-loop vacuum energy sequestered)}, \quad \text{while} \quad \partial \rho_{M1}/\partial G_{\mu\nu}^{phys} \neq 0$$
(small odd residual survives, gravitates).

Why this is the bottleneck. The two requirements pull against each other. Kaloper–Padilla sequestering removes the matter-loop vacuum energy globally but, by construction, leaves an *arbitrary* finite, radiatively stable residual curvature — it does not say *which* residual, nor that the M1 odd residual specifically survives. Importing the external mechanism therefore does not fix the residual's value or M1's selective survival; that is an internal Z-Spin computation, not yet done. Until it is, three failure modes stay open **even if the odd invariant gives exactly 8.190**: (a) the residual is sequestered with the rest ($\rho_\Lambda \to 0$); (b) the global constraint reintroduces an arbitrary integration constant on top of the residual; (c) even–odd operator mixing under RG reintroduces a radiatively unstable additive term. Register: **Selective Sequestering Compatibility Theorem — OPEN** (the top A27 / A28 priority, ahead of the odd invariant).
Connected sub-problem: the doublet action and the exact-J_Z tension. A prerequisite is to write the odd sector as one consistent action and resolve the standing exact-J_Z tension: with exact J_Z and $\mu_- = 0$, a J_Z-invariant vacuum and a J_Z-odd operator give $\langle \Omega_- \rangle = 0$, so the linear source $q_-\Omega_-$ (§5, Theorem 2) and the quadratic cross-coupled invariant (§5.3) are not yet the same action. The needed object is the explicit doublet action

$$S_- = \int\sqrt{-g}\,[\,-\tfrac{1}{2}\, q_a (K_q^{-1})_{ab} q_b + \bar{M}_P^2\, q_a \Omega_a\,],$$

together with a proof that the kernel K_q here is the M33-transplanted $K_-^{vac} = P_- X^\dagger X\, P_-$ (§5.3), and the computation of its rank and determinant-line normalization. Register: **Odd Doublet-Action / Kernel-Identification — OPEN**.

Methodological note (v2.3): provenance, and the limits of internal consistency.
Provenance. A substantial part of the channel-(ii) material — the R0 sequestering framing (§4), the transmutation computation (§7.1), the gauge cross-check and trunk-first ordering (§8.1), and verification checks 13–18 — originated in **AI-assisted deep-search exploration** and has been incorporated here, in

places nearly verbatim. We state the epistemic consequence plainly: the internal consistency of these sections, and the fact that the verification suite reproduces their numbers, is **not independent validation**. It is in part the analysis re-reading its own output — a documented failure mode in which an assistant's critique becomes the paper's self-description. The contributions appear correct and are anchored to external results (Kaloper–Padilla is published; the transmutation reproduces the corpus's own DERIVED v; the gauge cross-check agrees with ZS-A22 B2), so this is *not* fabrication — but it is a lack of independence. The open theorems above (§8.1 Odd Determinant–Torsion Identification; §8.2 Selective Sequestering Compatibility; the Odd Doublet-Action) are exactly the points where internal consistency cannot substitute for external checking. Breaking this loop requires **external input** — review by a noncommutative / spectral-geometry specialist (in particular on the even-dimension torsion-vanishing question, §8.1c) and confrontation with data (DESI for channel iii) — not a further internally-consistent revision. v2.3 records this rather than obscuring it.

Channel (iii), at equal weight. Channel (iii) — the observed Hubble scale realized as dynamical dark energy — deserves equal standing with channel (ii), not fallback status. The dimensional estimate $\Lambda_{QCD}{}^3H_0 \approx 6$ meV lands within a factor ~3 of the observed 2.24 meV — closer, in raw scale, than the channel-(ii) blind benchmark (90× short) — and unlike channel (ii) it is **testable now**: DESI DR2's 2.8–4.2σ preference for evolving dark energy [9,10] is precisely the external datum that discriminates it. The standing caveats hold (the scaling $\rho \propto \Lambda^3H$ is assumed, the ghost realization has $c_s^2 < 0$, the estimate is observation-anchored not blind), so channel (iii) is HYPOTHESIS like channel (ii); the point is that the betting order between (ii) and (iii) is genuinely open, and (iii) carries the near-term external test.

§9. Pre-registered falsification gates

G1 (C1′). If the heat-trace factorization is shown to fail for the corpus's actual D_Z (e.g. a non-product Dirac coupling is forced), Theorem 1 collapses. **G2 (M1-(ii)).** If no Z_2-odd dimension-4 electroweak operator can be built from the master action ($O_{EW}{}^{(-)} \equiv 0$), the four-form residual vanishes and M1 is dead. **G3 (M1-(iii)).** If exact Z_2 does not force $\mu_- = 0$ — if a Z_2-invariant boundary state still admits $\mu_- \neq 0$ — the residual is not predicted. **G4 (M1-(iv)).** If a one-loop even–odd mixing term is generated with nonzero coefficient, the offset re-enters and sequestering fails. **G5 (S1).** If E_* is ever fixed by back-solving from the observed ρ_Λ, the result is REJECTED-COINCIDENCE by construction; only a blind derivation (scaling + Γ_0 from the action) counts.

§10. Non-claims

This paper does **not** claim B3 is closed, nor that it is provably impossible to close. It does **not** claim C1′ is metric emergence — it is a coupling theorem, DERIVED-CONDITIONAL on importing the metric. It does **not** claim M1 is derived: the four-form residual is HYPOTHESIS-strong, and its four conditions (§5.1) are OPEN; in particular condition (ii)'s odd-operator multiplicity is reframed in §5.3 (a selector-free quadratic invariant conditional on an uncomputed kernel rank), and $\mu_- = 0$ is not yet proved. It does **not** claim any value for E_*; both 9v and 3πv are REJECTED-COINCIDENCE. It does **not** assert the Zhitnitsky scaling over the Klinkhamer–Volovik scaling — which one applies is itself OPEN. And it does **not** revise anything upstream: ZS-A26, ZS-A17, and ZS-F23.4 are used as established; the half-line/Weyl-clock material and the retracted calculations are confined to Appendix B.

Three further non-claims were added in this version, retracting v1.7 over-reach. It does **not** claim $\Lambda_Z \approx$ 25.6 MeV as a prediction — it is back-solved from $\rho_{\Lambda,\text{obs}}$ and H_0, a diagnostic value. It does **not** claim the ghost $w_0 = -0.763$ as a confirmed match to DESI — a one-coordinate proximity is not a likelihood, and the model's w_a sign and sound speed are wrong. And it does **not** treat the dark-energy data as an input that supplies a branch; observation tests a completed branch.

A fourth non-claim. The §5.3 reformulation does **not** close B3-2. The determinant-line route is **NON-CLAIM** on parity grounds (the top exterior power $\psi_{-3} \wedge \psi_{-11}$ is J_Z-even, while M1 needs J_Z-odd). The viable cross-coupled route makes the residual a basis-invariant quadratic form, but the kernel $K_- = P_- X^\dagger X\ P_-$ borrows ZS-M33's joint operator from the RH-bridge / Weil-positivity context, an import this paper treats as **HYPOTHESIS** (a structural isomorphism, not a physical identity), with the deciding rank **uncomputed**. It also does **not** assert the geometric-carrier asymmetry as a selector. ZS-M28 Theorem 28.14 (χ_{-3} on the Y-icosahedron) is promoted to **DERIVED** by ZS-M30 Theorem 30.2/30.3 (the χ_{-3} Eisenstein face as the $t = 0$ endpoint of an action-derived Robin-Truncation family); with $\chi_{-11} \leftrightarrow$Q-register (ZS-M22, PROVEN) this grounds the doublet's two geometric legs — strengthening, but not closing, the cross-coupled reframing, since the ZS-M33 kernel transplant and its rank remain.

Pattern check (self-reference). The recurring failure mode of this corpus is "partial numerical agreement → whole-structure justification." Its forms have changed but not its logic: 3πv, 9v, π/A (tuned dimensionless combinations); then 2.335 TeV and 25.6 MeV (target inversions); then "w_0 is 0.9σ away, therefore the model fits." §7 names the discipline that ends it: **observation may reject or support a branch that the action has already completed; it may not be used to build one**. Promoting $w_0 = -0.763$ to DERIVED and declaring B3 **CLOSED** (as one external review urged) is the same pattern, and is declined — exactly as the v1.3 "certified-unclosable" and the ZS-A25 "CLOSED" over-reaches were.

§11. Conclusion

Stripped of the retracted scaffolding, the absolute vacuum scale B3 has a clean four-root structure {C1′, R0, M1, S1}, and — the result of this version — a single structural reason it stays open. **C1′** closes the metric coupling, conditional on importing the metric, by the standard almost-commutative product whose register half the corpus already proved (Theorem 1). **R0** (even sequestering) and **M1** (a positive, constant topological four-form residual, Theorem 2) are well-motivated programmes — M1 has external precedent (Urban–Zhitnitsky; Klinkhamer–Volovik) — but each rests on unmet conditions: R0 must actually remove c_0, and M1 needs a derived Z_2-odd electroweak operator, a proved $\mu_- = 0$, and zero even–odd mixing. The mass-gap language is corrected: ZS-Q7 supplies an exact *dimensionless Liouvillian relaxation* gap, not a physical mass gap, and local-particle decoupling does not by itself exclude M1 — neither is "answered."

S1, the blind scale, is the deepest gate, and §7 says why: the A–Q-only inputs (dimensionless invariants of A and Q) do not by themselves fix the dimensionful cosmological constant (Buckingham-π). This is the **restricted A–Q-only no-go (DERIVED)**. The 10^{-121} hierarchy's origin is then organized by a *non-exhaustive* taxonomy of known routes — a tuned exponent (numerology, barred), an action-fixed transmutation (channel ii, computed in §7.1 but not closed), or the observed Hubble scale (channel iii, dynamical dark energy) — and this both explains the A22–A27 history and disciplines what may be claimed.

On the dark-energy lever. v1.7 over-reached here, and the internal v1.8 audit withdrew it: $\Lambda_Z \approx 25.6$ MeV was back-solved from $\rho_{\Lambda,\text{obs}}$ and H_0 (not predicted); the two scalings $\rho \propto \Lambda^3 H$ and $\rho \propto \Lambda^8/\bar{M}_P^4$

cannot be mixed; and the ghost $w_0 = -0.763$ is a toy discriminator (assumed scaling, one-coordinate, w_a sign wrong, $c_s^2 < 0$), not a confirmed prediction. Dark-energy data *test* a completed branch; they do not supply one.

B3 remains OPEN — not closed, and not impossible. Declaring it CLOSED on a back-solved scale plus a one-coordinate w_0 match would be the old numerology in a new form, and is declined. What is now *settled* is the reason B3 is open (the restricted A–Q-only no-go) and a *non-exhaustive* classification of the known routes to closing it. The constructive programme — $H_{J=-}^{0,4} \to G_- \to g_- \to \Lambda_- \to \chi_- \to \rho_\Lambda^{pred}$, under the blind protocol of §9 — is the work of a separate paper (ZS-A28); A27 closes here as the classification of the obstruction. The discipline is unchanged: E_* and Λ_- are never back-solved, $\mu_- = 0$ is proved not declared, and $O_{EW}^{(-)}$ is derived not chosen. With all of these and a scale from real dynamics, B3 would close honestly; without them, it does not.

Acknowledgements and Code Availability

This is the public release (v2.3). Code availability — each script is a consistency / diagnostic suite (numpy / scipy / mpmath / sympy), **not a physical proof**: *verify_v14.py* (8 corrective checks); *verify_v16_routeA.py* (4 Route-A diagnostics); *verify_v17.py* (4 mass-gap + 9 v1.7 integration checks, MG5 demoted); *verify_v18.py* (Dimensional-Closure DC0–DC3 + v1.8 retractions); and *verify_v21.py* (the §5.3 multiplicity-2 reformulation audit — parity, basis-covariance, and rank checks), and *verify_v23.py* (the §7.1 channel-(ii) transmutation computation: the blind benchmark, the post-unsealing 112/120 comparison, the back-solved C_odd target, and the gauge cross-check — all labelled NOT a closure proof). The corrective checks supersede, and are not added to, the historical counts of v1.1–v1.3 (Appendix B).

Appendix A — Diagnostic and Consistency Checks

These are reproduced by *verify_v14.py* (corrective), *verify_v17.py* (mass-gap and the v1.7 corrections), *verify_v16_routeA.py* (Route-A), and *verify_v18.py* (the Dimensional Closure Theorem and the v1.8 retractions). **They are consistency and diagnostic checks, not physical proofs**: none computes an actual Z-Spin sequestering, odd gauge algebra, beta function, susceptibility, DESI likelihood, or radiative-stability result. The MG5 decoupling line was an interpretation, not a machine check, and is not counted.

A.1 Corrective audit (8/8) — confirms the retractions and the corrected map

#	Check	Result
A1	R3 toy: $q(4.5) \approx 0.32$ but $q(8,20,50) \to 0$	**R3 retracted ✓**
A2	constant Ohmic friction: $\int\gamma\, dt = \infty \Rightarrow q \to 0$	**freeze refuted ✓**
A3	Frobenius-normalized $K \Rightarrow (1/Q)\mathrm{Tr} = 1/11 \ \forall\ K$	**R5 tautology ✓**
A4	Weyl exp: $T^3 \to p = 3$, $T^4 \to p = 4$	**C1 = A17 only ✓**
C1p	$\mathrm{Tr}\, e^{-tD^2}$ factorizes; $p_{tot} = p_X$	**bridge exists ✓**
M1	$q_- = \chi_Z O/\bar{M}_P^2$; $\rho > 0$, constant	positive, no freeze ✓

#	Check	Result
M1s	$\rho \propto E_*^8/\bar{M}_P^4$ scaling	preserved ✓
S1	$E_* = v \Rightarrow \rho^{1/4}$ ~90× low	**OPEN ✓**

A.2 Mass-gap analysis (4/4) — the Yang–Mills framing of S1

#	Check	Result
MG1	needed $E_*/v \approx 9.50$; no clean constant matches (3π off 0.07)	**9v, 3πv REJECTED ✓**
MG2	mass gap = transmutation $\exp(-1/2b_0g^2)$; spans ~2000×	not integer×v ✓
MG3	block-Laplacian gap = A is a *Liouvillian relaxation* gap (dimensionless), not a Hamiltonian mass gap; no mass scale derived (Γ_0 is the undetermined S1)	**relaxation gap only ✓**
MG4	KV needs 2.3 TeV; Zhitnitsky Λ^3H needs only Λ_{QCD}	**TeV is scaling artifact ✓**

MG5 (the former "decoupling objection answered" line) is removed: it passed True without computing anything. Correct status — local-particle decoupling does not by itself exclude a topological M1 sector; M1's RG stability is OPEN.

A.3 Integration checks (9/9) — the v1.7 corrections (DE/ZH rows reinterpreted by A.5)

#	Check	Result
LG1	Q7 spectrum is a Liouvillian (decay-rate), not Hamiltonian (E_1-E_0)	**relaxation gap ≠ mass gap ✓**
R0	without even sequestering $\rho_{grav} = c_0 + \rho_{M1}$	**4th root needed ✓**
DE1	ghost $\rho_{DE}=\alpha H \rightarrow w_0 = -1+(1-\Omega)/(2-\Omega) = -0.763$, $w>-1$ always	**HISTORICAL / RETRACTED (A.5)**
DE2	vs DESI DR2 $w_0 = -0.82\pm0.065$	**HISTORICAL / RETRACTED (A.5)**
ZH1	Zhitnitsky $\Lambda_Z = (\rho/H)^{1/3} \approx 25.6$ MeV; $\Gamma_0 \approx 0.3$ GeV	**HISTORICAL / RETRACTED (A.5)**
FT1	KV $\rho\propto\Gamma_0^8$ (2.14×) vs Zhitnitsky $\rho\propto\Gamma_0^3$ (1.33×) per 10%	Zhitnitsky less tuned ✓
DT1	$\Lambda_- = \mu_0 \exp(-8\pi^2/b_0g^2)$ spans MeV–TeV with g	transmutation, not n×v ✓
TS1	$\chi_- = c_\chi\Lambda_-^4 \rightarrow \rho_\Lambda = C_-\Lambda_-^8/\bar{M}_P^4$	susceptibility form ✓
MG5→	former decoupling line acknowledged as hardcoded	**demoted from count ✓**

A.4 Route-A historical diagnostics (4/4) — documenting the superseded Weyl-clock (Appendix B.3)

These verify why the half-line / Weyl-clock route is a limited Route-A tool; they are diagnostic records of superseded material, not part of the 13 current checks.

#	Check	Result
WC1	$\mathrm{Tr}[A,B] = 0 \neq i\hbar M$: no finite canonical pair	**Weyl pair, not canonical ✓**
WC2	forward shift S: $S^\dagger S = I$, $SS^\dagger = I - \lvert 0\rangle\langle 0\rvert$	**unilateral / number-phase ✓**
WC3	$\pi/A \approx 39.2$, $2\pi/A \approx 78.5$: non-integer, ~120 orders < 10^{121}	**$\pi/A \neq T_{max}$ ✓**

#	Check	Result
WC4	$\Lambda_n = (2\pi n+\alpha)\hbar/T_{cyc}$: branch equivalence	**central shift returns; O2 not O1 ✓**

A.5 Dimensional Closure and the v1.8 retractions (7/7)

#	Check	Result
DC0	locked inputs all dimensionless; $\rho_\Lambda/\bar{M}_P^4 \approx 7\times10^{-121}$	**Buckingham-π: scale must be imported ✓**
DC1	$A^n=10^{-121} \to n=109.6$; $(2A/Q)^n \to n=65.4$	**tuned exponent = numerology ✓**
DC2	transmutation to 2.3 TeV needs $g\approx0.57$; finite register has no β	**Exhaustion (c): new sector ✓**
DC3	$\rho\sim\Lambda^3H$; $H/\bar{M}_P\approx6\times10^{-61}$, no tuned exponent	**dynamical DE, w≠−1 ✓**
RET-ZH	$\Lambda_Z=(\rho_{obs}/H_0)^{1/3}$=25.6 MeV: inputs ARE the target	**back-solve, not prediction ✓**
RET-MIX	Λ_Z in $\rho=C\_\Lambda^8/\bar{M}_P^4 \to \rho^{1/4}\sim2.7\times10^{-13}$ eV (10^{10} low)	**two scalings cannot mix ✓**
RET-DE	ghost $w_0=-0.763$: assumed scaling, 1-coord, w_a wrong, $c_s^2<0$	**toy discriminator, not a fit ✓**

Appendix B — Correction History (v1.0–v1.4)

The development that produced this map is retained here for the record. The main body does not depend on it.

B.1 The three retracted calculations

C1 (RETRACTED). The register→metric "no-go" ($p = 0$ vs $p = 3$) only re-proved ZS-A17 and conflated spatial $p = 3$ with spacetime $p = 4$; $\Sigma(\lambda_n)^0 = N$ was a definition. Replaced by Theorem 1 (C1′). **R3 + Theorem A27.13 (RETRACTED).** The Schwinger–Keldysh "freeze" was a finite-time snapshot; the v1.3 Caldeira–Leggett repair gives a constant Ohmic friction, which yields $\int\gamma\, dt = \infty$ and $q \to 0$ (it does not freeze). Replaced by Theorem 2 (M1). **R5 (RETRACTED).** The blind coefficient returned 1/Q only because the kick operator was Frobenius-normalized; it was the input, not a prediction. And **"O2 closed"** was downgraded to "a representation template that can close O2."

B.2 Version line

Version	Date	What it was
v1.0	Jun 2026	Half-line conjugate eigenvalue terminus (Route A); 18 checks.
v1.1	Jun 2026	Finite-register dissolution + sequestered electroweak kick (Route B).
v1.2	Jun 2026	Ran C1/R3/R5 and the Z-Telomere Weyl clock; claimed two roots + "O2 closed."
v1.3	Jun 2026	Closure attempt (Caldeira–Leggett, Jacobson); over-reached to "certified-unclosable."
v1.4	Jun 2026	Major audit: RETRACTED C1, R3, A27.13, R5; re-decomposed to {C1′, M1, S1}.
v1.5	Jun 2026	Clean restatement: corrected map as main body, mass-gap framing of S1.
v1.6	Jun 2026	Restored the Route-A Weyl-clock diagnostics (B.3) lost in the v1.5 compression.
v1.7	Jun 2026	4th root R0; Liouvillian-gap correction; w(z) lever; transmutation + susceptibility; A27.14 restored.

Version	Date	What it was
v1.8	Jun 2026	This paper: Dimensional Closure Theorem; retracts the v1.7 dark-energy over-reach; final classification version.

B.3 Route A: the Z-Telomere Weyl-Clock diagnostic (superseded; retained for the record)

The v1.0–v1.2 line tried to fix the absolute scale by replacing the continuous half-line representation $L^2([0,\infty))$ with a finite cyclic *event* algebra — the Z-Telomere Weyl Clock. The audit verdict is that this is a limited Route-A tool: it addresses boundary/branch selection (A26's O2) at best, not the absolute scale (O1). The diagnostics that establish this are retained here; they are reproduced by *verify_v16_routeA.py* (4/4) and are separate from the 13 current checks.

(a) The (1,0) obstruction is representation-specific (cf. §2, scope C). The deficiency index (1,0) of $L^2(\mathbb{R}_+)$, $\Lambda = -i\partial_T$ is an obstruction to that *naive Schrödinger* realization, not to all unimodular quantizations — path-integral, boundary-state, three-form, and global-constraint realizations exist, and relational or covariant-POVM clocks evade the Pauli objection. So "change the representation" is the right instinct.

(b) Discretization is only half a fix (WC2). Replacing T_4 by an event count $n \in \mathbb{N}$ removes $-i\partial_T$, but the forward shift $S|n\rangle = |n+1\rangle$ is a *unilateral* shift: $S^\dagger S = I$ while $SS^\dagger = I - |0\rangle\langle 0| \neq I$, so S is not unitary. The continuous half-line *momentum* problem becomes the *number–phase* problem. "Discretize and the self-adjoint problem vanishes" is false.

(c) The genuine advance: a Weyl pair, not a canonical pair (WC1). No finite matrices satisfy $[\Lambda, T_4] = i\hbar I$ — taking traces gives $\mathrm{Tr}[A,B] = 0 \neq i\hbar M$. The correct finite structure is the clock–shift *Weyl* pair $U|n\rangle = |n+1 \bmod M\rangle$, $V|n\rangle = e^{2\pi i n/M}|n\rangle$, $UV = e^{-2\pi i/M}VU$, with boundary condition $U^M = e^{i\alpha}I$. All operators are finite and unitary, so there is no deficiency-index family. This is the part of the idea worth keeping.

(d) α = π/2 is a candidate, not a theorem. F18's quarter-turn (the relative phase of two local rotation generators) and the F10 handshake motivate $U^M = iI \Rightarrow \alpha = \pi/2$. But F18's π/2 is *local* while A27's α is the *global* T_4 boundary holonomy; identifying them needs a Commutator–Four-Volume Holonomy Theorem, $\Phi_{vol}([J_{R1}, J_{R2}]) = \mathrm{Hol}_{\partial M}(A_3)$. Without it, $\alpha = \pi/2$ is HYPOTHESIS-strong (30–60%).

(e) π/A is not the four-volume scale (WC3). $\pi/A \approx 39.2$ (a phase-effective handshake budget) and $2\pi/A \approx 78.5$ (a 2π completion count) are *non-integer* and ~120 orders below the ~10^{121} four-volume cell count. A finite cyclic capacity M must be an integer encoding 10^{121}; π/A cannot be M, and rounding it to the nearest integer after seeing the target is numerology. (dim Y = 6 is the per-cell channel count, not the universe-total Y information.)

(f) The decisive diagnostic: Floquet quasi-energy returns A26's central shift (WC4). Folding the four-volume clock modulo M makes Λ a Floquet *quasi-energy*: $\Lambda_n = (2\pi n + \alpha)\hbar/T_{cyc}$, with branch equivalence $\Lambda \sim \Lambda + 2\pi\hbar/T_{cyc}$. The clock fixes the spacing and the branch twist α, but the absolute zero is defined only mod $2\pi\hbar/T_{cyc}$ — A26's central shift reappears as a quasi-energy branch label. **So the cyclic clock can close O2 (branch selection), not O1 (absolute scale)**, which is exactly why the current main body is C1′ / M1 / S1 and not the clock.

(g) The Route-A theorem ladder (all OPEN except the holonomy lock). T1 Event-Volume Emergence — $T_4(n) = \Sigma_{j<n}\, \Delta V_4(j)$ from the action (OPEN; this "Volume Emergence Bridge" is the same metric-import gap that Theorem 1/C1′ resolves *conditionally*). T2 Telomere Weyl Algebra — $H_{cyc} \cong \mathbb{C}^M$, $UV = e^{-2\pi i/M}VU$ from the BV–BFV boundary symplectic form (OPEN). T3 Seam-Holonomy Lock — $U^M = iI \Rightarrow \alpha = \pi/2$ (HYPOTHESIS-strong). T4 Telomere Capacity — M or T_{cyc} from A, Q, and the action,

with $M \neq \pi/A$ kept open (OPEN). T5 Radiative Invariance — $\partial \Delta V_{vac}/\partial g_{\mu\nu} = 0$ (OPEN). Verdict: the half-line → Weyl-clock replacement may close the self-adjoint-domain and boundary-phase problems, but the metric four-volume scale and radiative stability remain separate OPEN gates — the same gates the main body names S1 and M1-(iv).

Appendix C — Deep-Exploration Record (mass-gap cycle)

Step 0 (long list). Routes to supply S1's scale: (a) $E_* =$ (integer)$\cdot$v; (b) dimensional transmutation of the Z-Spin coupling; (c) the finite block-Laplacian gap $A\cdot\Gamma_0$; (d) Klinkhamer–Volovik scaling $E_*^8/\bar{M}_P^4$; (e) Zhitnitsky scaling $\Lambda^3 H$; (f) a thermal spectral moment rising to TeV; (g) back-solve from ρ_Λ. *Dropped:* (a) and (g) (numerology / circular). *Retained:* (b)–(f) as the structure of the mass-gap question.
Steps 1–3 (issues + status). I1 reject the integer-multiple shortcut — **done** (MG1/MG2, REJECTED-COINCIDENCE). I2 is the Z-Spin gap a physical mass gap — **no**: it is an exact *dimensionless Liouvillian relaxation* gap (= A), not a Hamiltonian mass gap (MG3, corrected in §5.2). I3 what sets the dimensionful scale — **OPEN**, and §7 shows it must be imported (the restricted A–Q-only no-go). I4 which scaling governs ρ_Λ (KV vs Zhitnitsky) — **OPEN**, to be derived from a common action, not chosen by observation (MG4; corrected in §6.4). I5 decoupling — **not answered**: local-particle decoupling does not by itself exclude M1, but M1's RG stability is OPEN (former MG5, now demoted).
Step 4 (convergence). The exploration converged on a sharper S1: the gap is computable, the open content is (Γ_0, scaling), and the TeV scale is conditional on the KV scaling. No node required re-splitting. **CONVERGENT; B3 still OPEN. Step 5 (value + self-reference guard).** Value: S1 is reframed from "missing 2.3 TeV" to a well-posed action-level question, and M1 gains external precedent. **Anti-numerology: NOT APPLICABLE** (no value asserted; 9v, 3πv REJECTED). **Self-reference guard:** the temptation was to declare $E_* = 9$v a "geometric" closure; the mass-gap mechanism itself forbids it (a gap is not an integer multiple). The idea is the right key precisely because it rules out the shortcut it superficially suggests.

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[ZS] Internal: ZS-A17 (Spin–Metric Independence), ZS-A26 (central-shift obstruction), ZS-Q7 (master-equation spectrum $\lambda(\lambda+2A/Q)(\lambda+A)=0$), ZS-Q11 ($A_{ZS} = M_3 \oplus \mathbb{C} \oplus M_5$), ZS-F23.4 (trace-preserving central embedding).

Version History

v2.3 (June 2026): Conservative correction integrating two external-review documents. (1) §7.1 status downgraded: the Factorized-Determinant formula is DERIVED from PROVEN spectral inputs (not "PROVEN machinery"); the blind benchmark (2.48×10^{-5} eV) is computed but the M1 mechanism is *unvalidated*, and "112/120 orders" is a post-unsealing comparison, not a blind output or independent evidence; C_{odd} = 8.190 is a back-solved falsification *target*, not a prediction; the code/text Z-sector scale is reconciled (range 10^{-64}–10^{-23} GeV, ~40–80 orders below Planck — v2.2's "10^{-47} / 47 orders" was inconsistent with the verification); the "non-integer = numerology" criterion is corrected to "is the weight fixed *a priori* by the spectrum / measure?". (2) The Ray–Singer / Cheeger–Müller identification is demoted from DERIVED-CONDITIONAL to **HYPOTHESIS** and registered as the Odd Determinant–Torsion Identification Theorem (OPEN): a single coexact determinant is not the full RS torsion, the odd cohomology is non-acyclic (determinant-line normalization needed), and — sharpest — RS torsion is *trivial* for even-dimensional closed orientable manifolds with unitary coefficients (the Y-sector is 6-dim, even), so the route may vanish unless the non-unitary conductor data evades it — an external NCG question. (3) §8.1 over-framing **withdrawn**: "wall = one number" / "exponent 34.58 → B3 closes DERIVED" → the odd invariant is one necessary-not-sufficient S1 subgate; B3 closes only with all of B3-1…B3-7 (R0 trunk + remaining M1 conditions + coefficient + scaling). (4) The **Selective Sequestering Compatibility Theorem** (R0+M1 in one action; $\partial\Delta V/\partial G = 0$ while $\partial\rho_{M1}/\partial G \neq 0$) is registered as the TOP open problem (§8.2), ahead of the odd invariant, with the connected Odd Doublet-Action / exact-J_Z tension; channel (iii) ($\Lambda_{QCD}{}^3H_0 \approx 6$ meV, DESI-testable) is given equal weight. (5) A **methodological note** records that the channel-(ii) analysis originated in AI-assisted deep-search and that internal consistency is not independent validation — the loop is broken only by external review (NCG specialist + DESI), not a more faithful revision. The conclusion's v2.1 reversion ("Dimensional Closure Theorem," "exactly three channels," "corpus is dimensionless") is fixed and version labels unified. **B3 remains OPEN**. No new fitted parameters. ($A = 35/437$, $Q = 11$, $\dim Z = 2$) LOCKED.

v2.2 (June 2026): Computes channel (ii). Adds §7.1, the action-fixed transmutation computation §7 left as the one live no-observation route. Riding the corpus's PROVEN Higgs-VEV transmutation (Factorized Determinant Theorem, $v = \bar{M}_P \exp[-(38/9)(11 \ln 2 + \ln 3)] = 245.93$ GeV) **blind** gives $\rho_\Lambda^{1/4} = 2.48\times10^{-5}$ eV — recovering **112 of the 120 orders** the additive cutoff^4 estimate misses, validating the topological-transmutation *frame* as directionally correct (HYPOTHESIS-strong; the 112 orders are forced, not fit). But it falls **90× short**, the closing $\times 9.5 = E_*/v$ being exactly the documented S1 trap (3π off 0.07, or $2\nu_{now}\pi/A$ with the forbidden epoch $\nu_{now} = 3.526$, or $32\pi^2/(b_Z A)$ with the underived $b_Z = 14.25$). The even-sector invariant was a proxy; M1 is Z_2-odd, so closure needs the **odd**-sector spectral log-determinant to force $C_{odd}^{sp} = 8.190$ (exponent 34.58) — and the natural integer odd invariants **bracket** observation ($10 \ln 2 + \ln 3 \rightarrow \times 3.9$ over; $12 \ln 2 \rightarrow \times 3$ under) without hitting it, so the match is neither forced nor numerology-free at present. The gauge-transmutation cross-check gives the *same exponent* ($b_0 g_-^2 = 2.283$), confirming the Z-sector does not confine ($\Lambda_- \approx 10^{-64}$–$10^{-23}$ GeV, reproducing ZS-A22 B2) and that the Higgs / Factorized-Determinant route is the better-grounded one. **§8.1 framing**: v2.2 framed the A28 task as a single Ray–Singer-torsion computation. *Corrected in v2.3* to one necessary-not-sufficient S1 subgate, with the torsion identification demoted to a HYPOTHESIS (even-dimension vanishing concern). References [13] (asymptotic-freedom transmutation) and [14] (Cheeger–Müller) added. **B3 remains OPEN** — frame validated, wall = one number. No new fitted parameters. ($A = 35/437$, $Q = 11$, $\dim Z = 2$) LOCKED.

v2.1 (June 2026): Integrates an external technical review. Three substantive corrections. (1) The former "Dimensional Closure Theorem" (§7) is **restricted** to an A–Q-Only Scale-Generation No-Go: A, Q and the dimensionless invariants alone cannot fix the dimensionful ρ_Λ, but the corpus is not "purely dimensionless" ($\bar{M}_P$, v, Λ_{QCD} are dimensionful), so B3 is a hierarchy/selector problem; the three "channels" become a non-exhaustive taxonomy, and an *action-fixed* transmutation exponent is now correctly classed as legitimate nonperturbative physics (only a *target-fit* exponent is numerology). (2) "B3 openness proven" is corrected to *A–Q-only inputs underdetermine B3* (an absolute no-go is not established). (3) §5.3 is corrected: the parity statement now reads "even-degree objects are J_Z-even" (not "only linear objects are odd"); the quadratic invariant's basis-independence is stated conditional on K_- transforming covariantly; and the Choi–Stinespring / Kraus elevation is withdrawn (the rank is basis-independent linear algebra, but the CPTP-channel interpretation is OPEN). **One positive addition**: R0 gains an external **Kaloper–Padilla sequestering** lift (§4, a four-form / 4-volume mechanism with no new local DOF), promoting R0 to DERIVED-CONDITIONAL on identifying the corpus Z_2-even four-form with the sequestering global constraint; the closure programme (§8.1) is reordered to attack R0 (trunk) before the M1 / scale terminals. **Hygiene**: stale version labels (§1, §6.4) updated; the §8.1 B3-2 gate rewritten to the rank-test form; Appendix A.2 (MG3) and A.3 (DE1/DE2/ZH1) marked relaxation-gap-only / HISTORICAL-RETRACTED; code availability enumerated; verification reissued as *verify_v21.py* with the conductor and M30 checks demoted to corpus citations. **B3 remains OPEN**; no new fitted parameters. ($A = 35/437$, $Q = 11$, $\dim Z = 2$) LOCKED.

v2.0 (June 2026): Initial public release. Consolidates the internal v1.0–v1.10 research line into a single document. The four-root map **{C1′, R0, M1, S1}** (§§3–6), the **Dimensional Closure Theorem** (§7, DERIVED), and the **M1 multiplicity-2 reformulation** (§5.3: the determinant-line route is NON-CLAIM on parity grounds; the cross-coupled quadratic invariant $\rho_\Lambda = \frac{1}{2}\,\bar{M}_P^{-4}\,\Omega_-^T K_- \Omega_-$ is the viable reframing; B3-2 is recast as a finite rank test on the ZS-M33 Reading-C kernel; the χ_{-3} / χ_{-11} geometric legs are DERIVED via ZS-M30 Thm 30.2/30.3 and PROVEN via ZS-M22) are integrated. All retractions are preserved (Appendix B: the half-line / Weyl-clock calculations; the v1.7 dark-energy lever, Appendix A.5). **B3 remains OPEN** — its openness *proven* by §7, every route to closure classified, and a CLOSED declaration explicitly declined; the full constructive closure (rank computation, $\mu_- = 0$, even-sequestering, blind ρ_Λ) is deferred to ZS-A28. No new fitted parameters beyond the consolidated record; 3πv, 9v, π/A, and the back-solved TeV/MeV targets remain REJECTED-COINCIDENCE. (A = 35/437, Q = 11, dim Z = 2) LOCKED.

v1.10 (June 2026): Corrects one corpus-coverage error in v1.9. v1.9 §5.3 / §10 recorded that no ZS-M30 existed in the working corpus and therefore kept the χ_{-3} geometric carrier at DERIVED-CANDIDATE. ZS-M30 v1.0 ("Z-Spin Duality and the RH Bridge") was subsequently supplied; its **Theorem 30.3** (Robin-Truncation Boundary Family, **DERIVED**, from the non-minimal coupling $F(\Phi) = 1 + A|\Phi|^2$, ZS-F0 §8.2 Theorem 8.1 PROVEN) explicitly promotes ZS-M28 Theorem 28.14 from **DERIVED-CANDIDATE to DERIVED** by locating the χ_{-3} Eisenstein carrier as the t = 0 endpoint of an action-derived Robin family. §5.3 and §10 are corrected accordingly: the doublet's two geometric legs (χ_{-3}↔Y-icosahedron, DERIVED via M30; χ_{-11}↔Q-register, PROVEN via M22) are corpus-grounded, strengthening the cross-coupled reframing's geometric motivation. **The core results are unchanged**: the determinant-line route stays **NON-CLAIM** (ZS-M30 is orthogonal to the J_Z-parity obstruction — it carries no wedge/parity content), the cross-coupled quadratic invariant stays the viable reframing, and **B3 stays OPEN** (the ZS-M33 kernel transplant into A27 and its rank remain uncomputed; R0 / S1 untouched). One minor internal inconsistency in ZS-M30 is noted: §5.3(c) still labels the Robin interpolation OPEN (O-M30.2) while §5.5 / Step 8 closes it to DERIVED; we adopt the latter, considered status. No new fitted parameters. (A = 35/437, Q = 11, dim Z = 2) LOCKED.

v1.9 (June 2026): Adds §5.3, sharpening condition (ii) of the M1 programme. **Corrects the record**: the J_Z-odd dimension-4 BRST cohomology realized in ZS-M27 is *two*-dimensional (the ramified-prime characters χ_{-3}, χ_{-11}), not empty — the earlier "odd = 0" reading confused the non-isomorphic seam-grading (4,0) of ZS-M28.12 with the Clifford grading; condition (ii) is a genuine **multiplicity-2**. Audits two selector-free reformulations: the **determinant-line** path is **NON-CLAIM** (the top exterior power $\psi_{-3} \wedge \psi_{-11}$ is J_Z-even while M1 needs J_Z-odd; the exchange involution is a different $\mathbb{Z}_2$ from the seam), while the **cross-coupled quadratic invariant** $\rho_\Lambda = \frac{1}{2}\,\bar{M}_P^{-4}\,\Omega_-^T K_- \Omega_-$ is the viable reframing (basis-invariant; rank a CP/Kraus invariant by Choi–Stinespring), converting B3-2 into a finite **rank test** on the ZS-M33 Reading-C kernel $K_- = P_- X^\dagger X\, P_-$. Honest ledger: the kernel transplant from the RH-bridge context is **HYPOTHESIS**, the rank is uncomputed, and R0 / S1 are untouched, so **B3 remains OPEN** — multiplicity-2 is reframed, not closed. No new fitted parameters (dim $\Lambda^2\mathbb{C}^2 = 1$ and the rank classification are standard). (A = 35/437, Q = 11, dim Z = 2) LOCKED.

v1.8 (June 2026): Final classification version. Adds the **Dimensional Closure Theorem** (§7, DERIVED): the locked corpus is dimensionless, so by Buckingham-π it cannot fix the dimensionful ρ_Λ; the 10^{-121} hierarchy forces the missing scale's origin into three known (non-exhaustive) channels — a tuned exponent (numerology, barred), a new-sector transmutation (Exhaustion case (c)), or the observed Hubble scale (dynamical dark energy). **Retracts the v1.7 dark-energy over-reach** (§6.4, Appendix A.5): $\Lambda_Z \approx 25.6$ MeV was back-solved from $\rho_{\Lambda,obs}$ and H_0 (a diagnostic, not a prediction); the two scalings $\rho \propto \Lambda^3 H$ and $\rho \propto \Lambda^8/\bar{M}_P^4$ cannot be mixed (the v1.7 chain fails by $\sim 10^{10}$); and the ghost $w_0 = -0.763$ is a toy discriminator (assumed scaling, one-coordinate, wrong w_a sign, $c_s^2 < 0$), not a confirmed prediction.

Corrects internal contradictions (§6.2, Conclusion, Appendix C now state the relaxation gap is not a mass gap and decoupling is not “answered”); renames Appendix A to *Diagnostic and Consistency Checks* and reorders A.3/A.4; fixes Reference [10] (Gu et al., not Lodha). The constructive closure programme is deferred to ZS-A28. B3 remains OPEN — now with its openness *proven* (a dimensional no-go) and every route classified; a CLOSED declaration is explicitly declined.

v1.7 (June 2026): Integrates three external reviews. Corrections: the Q7 gap is identified as a *Liouvillian relaxation* gap, not a Hamiltonian mass gap (§5.2); the map gains a fourth root **R0** (even-sector sequestering of the bare offset, §4), since M1's residual otherwise sits on c_0; the former MG5 decoupling check is demoted (it was hardcoded); and “zero free parameters” becomes “zero *fitted* parameters, candidate coefficients OPEN.” Constructive additions: the KV–Zhitnitsky fork is recast as the *same* M1 four-form realized constant-vs-H-tracking and decided by the dark-energy equation of state — the linear ghost predicts $w_0 = -0.763$, within 0.9σ of DESI DR2 (§6.4); S1's scale is routed through dimensional transmutation (§6.5) and a topological susceptibility $\rho_\Lambda = C\_\Lambda\_^8/\bar{M}_P^4$ (§5.2); seven closure gates B3-1…B3-7 and a blind-prediction protocol are stated (§8.1); and the v1.4 emergent-gravity certification (Λ as an integration constant, Jacobson / entanglement-equilibrium) is restored and tied to the w(z) lever (§2.1). Editorial fixes: §1/verification references updated to v1.7, the §5.1 (ii)/(iii) glyph run-together repaired, new references [9–12]. B3 remains OPEN — now conditionally closable on a live measurement.

v1.6 (June 2026): Restores the Route-A / Z-Telomere Weyl-Clock diagnostics that the v1.5 restructure had compressed away, preserving them in the Correction History as Appendix B.3 (and four verifying checks as Appendix A.3). New record: the half-line (1,0) is representation-specific; discretization leaves a unilateral-shift / number-phase problem (so the fix is a Weyl pair, not a canonical pair); $\pi/A \approx 39.2$ is non-integer and ~120 orders below the 10^{121} four-volume, so it cannot be the cyclic capacity or T_{max}; and — decisively — folding the four-volume clock makes Λ a Floquet quasi-energy whose branch equivalence $\Lambda \sim \Lambda + 2\pi\hbar/T_{cyc}$ reintroduces A26's central shift, so the clock closes O2 (boundary/branch) but not O1 (absolute scale). The Route-A theorem ladder T1–T5 is recorded, with T1 (Event-Volume / Volume-Emergence Bridge) noted as the same metric-import gap that Theorem 1 (C1′) resolves conditionally. No change to the main body or to the 13 current checks; B3 remains OPEN.

v1.5 (June 2026): Clean restructured release. The corrected map (then stated as three roots C1′, M1, S1) is the main body; the half-line/Weyl-clock development and the retracted C1·R3·R5·A27.13 calculations are moved to Appendix B (Correction History). New: the precise conditional-no-go boundary (§2), the four explicit M1 conditions (§5.1), the Yang–Mills mass-gap analysis of S1 (§6) — rejecting $E_* = 9v / 3\pi v$ as REJECTED-COINCIDENCE, showing the Z-Spin gap is finite and computable while its scale reduces to Γ_0, and exhibiting the KV-vs-Zhitnitsky scaling fork — and the Vacuum-Residual Exhaustion Theorem target for any future absolute no-go (§8). Verification re-counted honestly as **v1.5 audit 13/13 PASS** (8 corrective + 5 mass-gap); the misleading “50/50” count is retired. Real Word heading styles and a table of contents added. New references [3–5] (Veneziano-ghost / q-theory vacuum energy), [8] (Yang–Mills mass-gap problem). **Verdict: B3 remains OPEN**. Zero new fitted parameters; (A = 35/437, Q = 11, dim Z = 2) LOCKED. Consolidated from internal Z-Spin Collaboration research notes through v1.4.